A Formal Calculus for Informal Equality with Binding

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Abstract

In informal mathematical usage we often reason about equalities involving binding of object-variables. We find ourselves writing assertions involving *meta-variables* and *capture-avoidance constraints* on where object-variables can and cannot occur free. Formalising such assertions is problematic because the standard logical frameworks cannot express capture-avoidance constraints directly.

In this paper we make the case for *extending* the logic of equality with meta-variables and capture-avoidance constraints. We use nominal techniques that allow for a direct formalisation of meta-level assertions, while remaining close to informal practice. We investigate proof-theoretical properties and we provide a sound and complete semantics in nominal sets.

1 Introduction

Perhaps *equality* is the simplest possible judgement form. Informal specification of logic and computation often involves equalities with *binding* and subject to conditions about *freshness*. For example:

λ-calculus:	$\lambda x.(tx) = t$	if $x \notin fv(t)$
First-order logic:	$\forall x.(\phi \supset \psi) = \phi \supset \forall x.\psi$	if $x \notin fv(\phi)$
π-calculus:	$\nu x.(P \mid Q) = P \mid \nu x.Q$	if $x \notin fv(P)$
Process algebra with data:	$\sum x.p = p$	if $x \notin fv(p)$

And for any binder $\zeta \in \{\lambda, \forall, \nu, \sum\}$:

Substitution:	$(\zeta y.u)[x \mapsto t] = \zeta y.(u[x \mapsto t])$	if $y \notin fv(t)$

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Here $fv(t)$ denotes the free variables of t. It is not hard to extend this short list with many more examples.

In the equalities above there are *two* levels of variable:

- x and y are variables of the system being axiomatised, we call these *object-level* variables.
- t, u, ϕ, ψ, P, Q and p range over terms of that system's syntax, we call them *meta-level* variables.

These equalities are subject to freshness side-conditions $x \notin fv(t)$ which make them something other than 'just equalities'.

A straightforward way to formalise these meta-level properties is to enrich to the logic of equality with meta-variables. In a setting with binders, we have to face the following problems:

- When is an object-variable fresh for a meta-variable? We only know this when the meta-variable is instantiated to a concrete formula or term (not mentioning meta-variables).
- What is suitable representation of α-conversion of object-variables in the setting with meta-variables?
- In the presence of meta-variables, substitution of terms for object-variables becomes non-trivial: what does it mean when we try to substitute a term for a variable in a meta-variable?

A number of different solutions have been proposed for these problems. The state of the art solution is the higher-order approach. This uses a hierarchy of types to model the difference between object- and meta-level variables [vB01, Mei92]. An alternative solution is to reject object-variables altogether, as is done in approaches using combinators [CF58, Bar84] and cylindric algebras [HMT85, LS04].

Unfortunately, none of these solutions allows for a *natural* formalisation. By this we mean a formalisation that permits a direct representation of the two-level structure which informal syntax displays in terms such as $\lambda x.t$ or $\forall x.\phi$. A consequence of this is that the formalisation of schemas of theorems and proofs are not always simple refinements but sometimes requires a fair amount of emulation. In De Bruijn's words [dB91]:

> "I think that in formalizing mathematics, and in particular in preparing mathematics for justification, it is usually elegant as well as efficient to do everything in the *natural* way."

There have been solutions to the problem of binding and meta-variables that embrace the difference between object- and meta-variables. One of these solutions uses so-called nominal techniques [GP02], which has two levels of variables, and a built-in notion of freshness of variables with respect to meta-variables.

Nominal techniques have been successfully applied to unification [UPG04], term rewriting [FG07] and first-order logic [Pit03]. The application was purely to represent and reason about formal *syntax* with meta-variables.

In this paper we explore **Nominal Algebra**, a novel but very natural application of nominal techniques to represent formal *logic* with meta-variables. In this way, and consistent with de Bruijn's philosophy, we allow particularly natural and yet fully formal representations of the kind of reasoning which is pervasively used in informal mathematical practice.

Nominal Algebra is a logic of equality which embraces the two-level variable structure by representing it directly in its syntax. Informal equivalences can be represented as axioms almost symbol-for-symbol. For example the equalities from the beginning of the Introduction are represented by:

λ-calculus:	$a\#X \vdash$	$\lambda[a](Xa) = X$
First-order logic:	$a\#X \vdash$	$\forall[a](X \supset Y) = X \supset \forall[a]Y$
π-calculus:	$a\#X \vdash$	$\nu[a](X \mid Y) = X \mid \nu[a]Y$
Proces algebra with data:	$a\#X \vdash$	$\sum[a]X = X$
Substitution:	$b\#X \vdash$	$(\zeta[b]Y)[a \mapsto X] = \zeta[b](Y[a \mapsto X])$

Here a and b are distinct *atoms* representing object-level variables; X and Y are *unknowns* representing meta-level variables; $[a]t$ is an abstraction of an atom a in a term t. Each equality is equipped with a *freshness condition* of the form $a\#X$ that guarantees that X can only be instantiated to a term for which a is fresh. The rest of this paper makes this formal.

We also develop the proof theory of nominal algebra and show that it supports the following key features of meta-level reasoning:

- instantiation of meta-variables, by means of *capturing* substitution of terms for meta-variables;
- α*-renaming* of object-variables and *capture-avoiding* substitution of terms for object-variables in the presence of meta-variables;
- generation of *fresh* object-variables inside a derivation.

Finally, we provide a denotation in so-called *nominal sets*. Nominal sets were introduced by Gabbay and Pitts in [GP02].[1] They have proved to be an effective model for syntax with names and binding (see for example [Pit03]). In fact, nominal sets have inspired the design of nominal terms, which form the basis of nominal algebra. For this reason, nominal sets permit a natural semantic interpretation of atoms a, abstractions $[a]t$, and freshness $a\#t$, which are not conveniently definable on 'ordinary' sets.

A conference version of this material was presented [GM07b]; in this paper the presentation is revised, extended, and the proof of completeness for a semantics

[1] In [GP02], nominal sets are called FM-sets, named after the Fraenkel and Mostowski who devised a permutation model of set theory in order to prove the independence of the axiom of choice from the other axioms of Zermelo-Fraenkel set theory with atoms [Bru96].

in nominal sets [GP02] is given in full detail. The authors have used nominal algebra to study capture-avoiding substitution for atoms [GM06a, GM07a], and first-order logic [GM06b, GM07c]. Also, nominal algebra served as the basis for nominal equational logic [CP07]; also a 'nominal algebraic' system, but making some different choices in presentation and design. We discuss this and other related work in the Conclusions.

Overview We introduce the syntax of nominal algebra in Section 2. In Section 3 we provide a notion of derivation of freshness and equality with the ability to impose axioms, and we provide some examples. In Section 4 we show a number of proof-theoretical results.

In Sections 5 to 7 we discuss the denotation in nominal sets. We give a brief overview of nominal sets in Section 5. In Section 6 we show how we can interpret freshness and equality in nominal sets, we define what constitutes a model of a theory (a signature with a set of axioms), and we show that our notion of derivability is sound with respect to the semantics. In Section 7 we show that derivability of equality (in some theory) is complete with respect to its models. We also show that derivability of freshness is not complete, but that validity of freshness can be expressed in terms of derivability of certain equalities.

We discuss related and future work in Section 8.

2 Syntax

We need a syntax in which expressions with meta-variables, such as $\lambda x.t$ and $x \notin fv(t)$, may be represented. We use nominal terms [UPG04] because they offer built-in support for meta-variables, abstraction, and freshness in a way that is close to informal practice.

2.1 Terms and Signatures

Definition 2.1. Fix a countably infinite collection of **atoms** $a, b, c, \ldots$ representing object-level variables. Fix a countably infinite collection of **unknowns** $X, Y, Z, \ldots$ representing meta-level variables. Fix **term-formers** f to each of which is associated some unique **arity** n which is a nonnegative number; write $\mathsf{f} : n$ to indicate that f has arity n. Assume these collections are disjoint.

Convention 2.2. We shall use a *permutative convention* that $a, b, c, \ldots$ range permutatively over atoms, so that for example a and b are always distinct.

In the rare circumstances where we do not want this behaviour, we choose another symbol such as a'.

For the purpose of α-conversion, we need to be able to rename atoms. We use *permutations* of atoms instead of the more common substitutions of atoms for atoms, because permutations have better mathematical properties; most notably,

permutations are capture-avoiding by definition (see the Introduction of [GP02] and [Pit03] for a detailed exposition).

Definition 2.3. Let $\mathbb{A} = \{a, b, c, \ldots\}$. A **permutation** π of atoms is a total bijection $\mathbb{A} \to \mathbb{A}$ with **finite support**, meaning that for some finite set of atoms (which may be empty) $\pi(a) \neq a$, *but* for all atoms not in that set, $\pi(a) = a$.

Finite support is a mathematical notion of 'most': π is a bijection on atoms such that $\pi(a) = a$ for *most* a.

We introduce some notation for permutations that we will need later on.

Definition 2.4. Write ι for the **identity permutation** such that $\iota(a) = a$ always. Write $\pi \circ \pi'$ for **functional composition** and write π^{-1} for **inverse**. This makes permutations into a group — write $\mathbb{P}$ for the set of all permutations. Write $(a\ b)$ for the permutation that **swaps** a and b, i.e. the permutation that maps a to b, b to a and all other c to themselves. Finally, write $a \in \pi$ when $\pi(a) \neq a$.

Using the above ingredients we can form terms.

Definition 2.5. Terms t, u, v are inductively defined by:

$$t \quad ::= \quad a \ \mid\ \pi \cdot X \ \mid\ [a]t \ \mid\ \mathsf{f}(t_1, \ldots, t_n)$$

We call $[a]t$ an **abstractor**; it uniformly represents the '$x.t$' or $x.\phi$' part of expressions such as '$\lambda x.t$' or '$\forall x.\phi$'. We call $\pi \cdot X$ a **moderated unknown**; it represents an unknown term on which a permutation of atoms is performed when it is instantiated. We write $\iota \cdot X$ just as X, for brevity.

In Section 3 we will see that in $\pi \cdot X$ the unknown X will get substituted for a term and then π will permute the atoms in that term. This notion is grounded in semantics [GP02] and permits a succinct treatment of α-renaming atoms (see Section 4.4 and [UPG04]).

Definition 2.6. A **signature** Σ is a set of term-formers with their arities.

Example 2.7. Here are some example signatures:

- $\{\mathsf{lam} : 1, \mathsf{app} : 2\}$ is a signature for the λ-calculus.

  We show how the terms in this signature relate to 'ordinary' syntax. For convenience identify atoms with *variable symbols*, then the syntax of the untyped λ-calculus is inductively defined by:

  $$e \quad ::= \quad a \ \mid\ \lambda a.e \ \mid\ ee$$

  The map -$'$ from untyped λ-term to nominal terms is inductively defined by:

  $$a' = a \qquad (\lambda a.e)' = \mathsf{lam}([a](e')) \qquad (e_1 e_2)' = \mathsf{app}(e_1', e_2')$$

  We generally sugar $\mathsf{lam}([a]t)$ to $\lambda[a]t$ and $\mathsf{app}(t, u)$ to tu.

- $\{\bot : 0, \supset\colon 2, \forall : 1, \approx\colon 2\}$ is a signature for first-order logic with equality (the symbol for equality inside the logic is $\approx$).

  We sugar $\bot()$ to $\bot$, $\supset(\phi, \psi)$ to $\phi \supset \psi$, $\forall([a]\phi)$ to $\forall[a]\phi$ and $\approx(t, u)$ to $t \approx u$.

When we define axioms for the λ-calculus and first-order logic, we shall extend these signatures with a term-former for representing capture-avoiding substitution.

Remark 2.8. Consistent with previous work on nominal rewriting [FG07] we do not impose an a priori sort system on terms. Although this allows us to write 'silly terms' like $\lambda(tu)$ and $\forall(t \approx u)$, it simplifies the presentation, and our results trivially specialise to the more specific cases.

Definition 2.9. Write $t \equiv u$ for **syntactic identity** of terms.

Note that if $\pi = \pi'$ then $\pi \cdot X \equiv \pi' \cdot X$, since permutations are represented by themselves. There is no quotient by abstraction so for example $[a]a \not\equiv [b]b$.

Syntactic identity $t \equiv u$ emphasises the difference from provable equality $t = u$, which is a logical assertion defined in Subsection 2.2, and object-level equality $t \approx u$, which is a term.

Definition 2.10. Say that a term t is **closed** when it does not contain any *unknowns*.

A closed term may still mention atoms, e.g. the terms a and $[a]b$ are closed and the terms X and $[a]X$ are not.

Definition 2.11. Write $a \in t$ for 'a **occurs in (the syntax of)** t', and $X \in t$ for 'X **occurs in (the syntax of)** t'. Similarly write $a \notin t$ and $X \notin t$ for 'does not occur in the syntax of t'.

Occurrence $a \in t$ is literal, e.g. $a \in [a]a$ and $a \in \pi \cdot X$ when $a \in \pi$.

2.2 Judgement Forms, Axioms and Theories

Definition 2.12. A **freshness** is a pair $a\#t$ of an atom a and a term t. Call a freshness $a\#X$ (so $t \equiv X$) **primitive**. Write Δ and ∇ for finite sets of *primitive* freshnesses and call them **freshness contexts**.

Intuitively we should read $a\#t$ as meaning '$a \notin fv(t)$' or in words 'a is fresh for t'. A reason this notion is quite subtle in nominal techniques is the unknowns X; $a\#X$ is *not* necessarily true even though $a \notin X$ *is* a fact of the syntax. An unknown X represents an unknown term in the syntax; $a\#X$ has the quality of a *promise* or *assertion* about what term that can be, or put another way, about what we can instantiate X to.

Definition 2.13. We may drop set brackets in sets of freshnesses, e.g. writing $a\#t, b\#u$ for $\{a\#t, b\#u\}$. Also, we may write $a\#t, u$ for $a\#t, a\#u$. Furthermore, for any set of freshnesses S write $a \in S$ when a occurs anywhere in S, and $X \in S$ when X occurs anywhere in S.

Definition 2.14. An **equality** is a pair $t = u$ where t and u are terms.

Equalities will be used to state that two terms are *provably equal*.

Definition 2.15. Nominal algebra has two *judgement forms*:

- A **freshness judgement form** $\Delta \vdash a\#t$ is a pair of a freshness context Δ and a freshness $a\#t$.
- An **equality judgement form** $\Delta \vdash t = u$ is a pair of a freshness context Δ and an equality $t = u$.

We may write $\emptyset \vdash a\#t$ as $\vdash a\#t$, and $\emptyset \vdash t = u$ as $\vdash t = u$.

Definition 2.16. A **theory** $\mathsf{T} = (\Sigma, Ax)$ is a pair of a signature Σ and a possibly infinite set of *equality* judgement forms Ax in that signature; we call them the **axioms**.

We do not allow freshness judgements as axioms; we shall see that they can be expressed using equalities instead (see Subsection 7.3).

Example 2.17. Here are some nominal algebra theories:

- LAM has signature $\{\mathsf{lam} : 1, \mathsf{app} : 2, \mathsf{sub} : 2\}$ and two axioms

$$\begin{array}{lll} (\beta) & \vdash & (\lambda[a]Y)X = Y[a \mapsto X] \\ (\eta) \quad a\#X & \vdash & \lambda[a](Xa) = X \end{array}$$

  where we sugar $\mathsf{sub}([a]t, u)$ to $t[a \mapsto u]$.

- FOL has signature $\{\bot : 0, \supset : 2, \forall : 1, \approx : 2, \mathsf{sub} : 2\}$ and seven axioms

$$\begin{array}{lll} (\mathbf{MP}) & \vdash \top \supset X & = X \\ (\mathbf{Mer}) & \vdash ((((X \supset Y) \supset (\neg Z \supset \neg W)) \supset Z) \supset V) & \\ & \qquad \supset ((V \supset X) \supset (W \supset X)) & = \top \\ (\mathbf{Qinst}) & \vdash \forall[a]X \supset X[a \mapsto Y] & = \top \\ (\mathbf{Qdist}) & \vdash \forall[a](X \wedge Y) \Leftrightarrow \forall[a]X \wedge \forall[a]Y & = \top \\ (\mathbf{Qextr}) \quad a\#X & \vdash \forall[a](X \supset Y) \Leftrightarrow X \supset \forall[a]Y & = \top \\ (\mathbf{Esubst}) & \vdash Z \approx Y \wedge X[a \mapsto Y] \supset X[a \mapsto Z] & = \top \\ (\mathbf{Erefl}) & \vdash X \approx X & = \top \end{array}$$

  Here we use standard classical logic sugar for $\top$, $\neg$, $\wedge$ and $\Leftrightarrow$.

  Axioms (**MP**) and (**Mer**) characterise propositional logic; axioms (**Qinst**), (**Qdist**) and (**Qextr**) characterise quantification; and axioms (**Esubst**) and (**Erefl**) characterise object-level equality.

  This axiomatisation is treated in detail in [GM06b, GM07c].

- SUB gives substitution term-former sub the correct behaviour in theories LAM and FOL. It is a family of theories, one for each signature Σ that includes sub, with axioms

$$\begin{array}{rlll}
(\mathbf{var}{\mapsto}) & & \vdash & a[a \mapsto X] = X \\
(\#{\mapsto}) & a\#Y & \vdash & Y[a \mapsto X] = Y \\
(\mathsf{f}{\mapsto}) & & \vdash & \mathsf{f}(Y_1, \ldots, Y_n)[a \mapsto X] = \mathsf{f}(Y_1[a \mapsto X], \ldots, Y_n[a \mapsto X]) \\
(\mathbf{abs}{\mapsto}) & b\#X & \vdash & ([b]Y)[a \mapsto X] = [b](Y[a \mapsto X]) \\
(\mathbf{ren}{\mapsto}) & b\#Y & \vdash & Y[a \mapsto b] = (b\ a) \cdot Y \\
(\eta{\mapsto}) & b\#X & \vdash & [a]\mathsf{sub}(X, a) = X
\end{array}$$

  For each term-former f (including sub), there is one axiom (f↦). Note the heavy use of freshness side-conditions to manage the relationship between atoms and unknowns.

  This axiomatisation is the topic of [GM06a, GM07a].

- CORE is a family of theories with no axioms; there is one such theory for each signature Σ. It has built-in α-equivalence, so for example $\lambda[a]a$ is equal to $\lambda[b]b$.[2]

  Theory CORE is discussed in Subsection 4.4.

Similar developments for other systems with binding, such as the process algebra with data [Gro97, Lut02, GMR+07] and the π-calculus [Par01] from the Introduction should also be possible.

3 A Derivation System

In this section we define notions of derivation which represent freshness assumptions on meta-variables (Figure 1), and permit axioms involving abstraction that are conditioned by freshness assumptions (Figure 2), just like we do in informal practice.

3.1 Permutation and Substitution Actions

Before we introduce our calculus, we elaborate on two important prequisites for the instantiation of axioms; we need to be able to *permute atoms in terms*, and *substitute terms for unknowns* in a capturing way.

Definition 3.1. The **(object-level) permutation action** $\pi \cdot t$ on terms is inductively defined by:

$$\pi \cdot a \equiv \pi(a) \qquad \pi \cdot (\pi' \cdot X) \equiv (\pi \circ \pi') \cdot X \qquad \pi \cdot [a]t \equiv [\pi(a)](\pi \cdot t)$$
$$\pi \cdot \mathsf{f}(t_1, \ldots, t_n) \equiv \mathsf{f}(\pi \cdot t_1, \ldots, \pi \cdot t_n)$$

[2] α-equivalence is expressed as a derivation rule: the (**perm**) rule from Figure 2. The (**perm**) rule is discussed in detail in Subsection 3.2.

Intuitively, π propagates through the structure of t until it reaches an atom or a moderated unknown. The reader should read the above definition as *syntactic sugar*. For instance, this means that $\pi \cdot [a](\pi' \cdot X)$ is *not* an actual term but sugar for $[\pi(a)]((\pi \circ \pi') \cdot X)$.

Composition and identity of permutations naturally extend to terms as shown in the following lemma.

Lemma 3.2. $(\pi \circ \pi') \cdot t \equiv \pi \cdot (\pi' \cdot t)$ *and* $\iota \cdot t \equiv t$.

Proof. By induction on the structure of t, using Definition 3.1. □

Substitution is the mechanism by which unknowns become terms, and is necessary in algebra in order to define instances of axioms.

Definition 3.3. A **(meta-level) substitution** σ is a function from unknowns to terms.

We write $[t_1/X_1, \ldots, t_n/X_n]$ for the substitution σ such that $\sigma(X_i) \equiv t_i$ for $1 \leq i \leq n$, and $\sigma(Y) \equiv Y$ for $Y \not\equiv X_i$. Write $[]$ for the **empty substitution** and write $\sigma \circ \sigma'$ for **composition of substitutions**, so $t[] \equiv t$ and $t(\sigma \circ \sigma') \equiv (t\sigma)\sigma'$.

Definition 3.4. The **(meta-level) substitution action** $t\sigma$ on terms is inductively defined by:

$$a\sigma \equiv a \qquad (\pi \cdot X)\sigma \equiv \pi \cdot \sigma(X) \qquad ([a]t)\sigma \equiv [a](t\sigma)$$
$$\mathsf{f}(t_1, \ldots, t_n)\sigma \equiv \mathsf{f}(t_1\sigma, \ldots, t_n\sigma)$$

Intuitively, σ propagates through the structure of t until it reaches an atom or a moderated unknown. σ 'evaporates' on an atom, and acts on the unknown X of a moderated unknown $\pi \cdot X$. The moderating permutation π then passes into the term substituted in that position. We suggest a reading of $\pi \cdot X$ as 'permute π in whatever X eventually becomes'.

Note that meta-level substitution does not avoid capture; $([a]X)[a/X] \equiv [a]a$. In this sense X is 'meta' and really does represent an unknown *term*. There is an exact and deliberate analogy here with context substitution, which is the substitution used when we write 'let - be a in $\lambda a.$-', to obtain $\lambda a.a$.

Definition 3.5. Give substitution and permutation actions higher precedence than abstraction and any of the sugared term-formers, and put substitution before permutation.

Note how substitution interacts with permutation in the case of an unknown, for example $((a\ b) \cdot X)[b/X] \equiv (a\ b) \cdot b \equiv a$. So π in X is 'waiting for a substitution to arrive', as also made formal in the following property:

Lemma 3.6. $\pi \cdot t\sigma \equiv (\pi \cdot t)\sigma$.

Proof. By induction on the structure of t, using Definitions 3.1 and 3.4. The case of $t \equiv \pi' \cdot X$ uses Lemma 3.2. □

Another permutation action is useful.

Definition 3.7. The **meta-level permutation action** t^π on terms t is inductively defined by:

$$a^\pi \equiv \pi(a) \qquad (\pi' \cdot X)^\pi \equiv \pi \circ \pi' \circ \pi^{-1} \cdot X \qquad ([a]t)^\pi \equiv [\pi(a)]t^\pi$$
$$\mathsf{f}(t_1, \ldots, t_n)^\pi \equiv \mathsf{f}({t_1}^\pi, \ldots, {t_n}^\pi)$$

Also for this permutation action, composition and identity of permutations extend to terms.

Lemma 3.8. *$t^{\pi \circ \pi'} \equiv t^{\pi' \pi}$ and $t^\iota \equiv t$.*

Proof. By induction on the structure of t, using Definition 3.7. □

In the presence of substitution, the two permutation actions $\pi \cdot t$ and t^π are interdefinable; however, sometimes one is more natural than the other, we shall point out how later (Remark 3.14).

Lemma 3.9. *Given a term t, let σ be a substitution that maps each $X \in t$ to $\pi \cdot X$, and let σ' be a substitution that maps each $X \in t$ to $\pi^{-1} \cdot X$.*
Then $\pi \cdot t \equiv t^\pi \sigma$ and $t^\pi \equiv (\pi \cdot t)\sigma'$.

Proof. By induction on the structure of t, using Definitions 3.1, 3.7 and 3.4 of $\pi \cdot t$, t^π and $t\sigma$. The only interesting case is when $t \equiv \pi' \cdot X$. Then we need to show $\pi \cdot (\pi' \cdot X) \equiv (\pi' \cdot X)^\pi \sigma$. Since $(\pi' \cdot X)^\pi \sigma \equiv (\pi \circ \pi' \circ \pi^{-1}) \cdot (\pi \cdot X)$ by Definitions 3.7 and 3.4, it suffices to show $\pi \cdot (\pi' \cdot X) \equiv (\pi \circ \pi' \circ \pi^{-1}) \cdot (\pi \cdot X)$. This follows using Definition 3.1 and the fact that $\pi \circ \pi' = \pi \circ \pi' \circ \pi^{-1} \circ \pi$.
The proof of $(\pi' \cdot X)^\pi \equiv (\pi \cdot (\pi' \cdot X))\sigma'$ follows similar lines. □

Definition 3.10. We extend notation for t^π, $\pi \cdot t$ and $t\sigma$ to freshness contexts Δ as follows:

$$\begin{array}{rcl} \Delta^\pi & \text{is} & \{\pi(a)\#X \mid a\#X \in \Delta\} \\ \pi \cdot \Delta & \text{is} & \{\pi(a)\#\pi \cdot X \mid a\#X \in \Delta\} \\ \Delta\sigma & \text{is} & \{a\#\sigma(X) \mid a\#X \in \Delta\} \end{array}$$

Note that Δ^π is a freshness context, but $\pi \cdot \Delta$ and $\Delta\sigma$ need not be.

3.2 Inference Rules

Definition 3.11. Define **derivability on freshnesses** (in some signature Σ) in natural deduction style by the rules in Fig. 1. Here we have used the following conventions:

- a and b range over *distinct* atoms (see Convention 2.2);
- π ranges over permutations;
- X ranges over unknowns;

$$\frac{}{a\#b}\,(\#\mathbf{ab}) \qquad \frac{\pi^{-1}(a)\#X}{a\#\pi\cdot X}\,(\#\mathbf{X}) \quad (\pi\neq\iota)$$

$$\frac{}{a\#[a]t}\,(\#[]\mathbf{a}) \qquad \frac{a\#t}{a\#[b]t}\,(\#[]\mathbf{b}) \qquad \frac{a\#t_1\ \cdots\ a\#t_n}{a\#\mathsf{f}(t_1,\ldots,t_n)}\,(\#\mathsf{f})$$

Figure 1: Derivation rules for freshness

- t and $t_1,\ldots,t_n$ range over terms;
- f ranges over term-formers; there is one copy of the rule for each term-former.

We use similar conventions henceforth.

Write $\Delta\vdash a\#t$ when a derivation of $a\#t$ exists according to the rules in Fig. 1 using the elements of Δ as assumptions; say that $a\#t$ **is derivable from** Δ. Write $\Delta\nvdash a\#t$ when $a\#t$ is not derivable from Δ.

Finally, we write $\Delta\vdash S$ for a set of freshnesses S when $\Delta\vdash a\#t$ for each $a\#t\in S$.

Note that the $(\#\mathbf{X})$ rule excludes the identity permutation ι. While there is no mathematical reason for this, there is a nice *computational* one: the algorithm obtained by reading the rules bottom-up, must terminate.

Example 3.12. In the signature of theory LAM (Example 2.17) we know $\vdash a\#\lambda[b]b$ and $a\#X\vdash a\#X(\lambda[a]Y)$:

$$\dfrac{\dfrac{\dfrac{}{a\#b}\,(\#\mathbf{ab})}{a\#[b]b}\,(\#[]\mathbf{b})}{a\#\lambda[b]b}\,(\#\mathsf{f}) \qquad \dfrac{a\#X \quad \dfrac{\dfrac{}{a\#[a]Y}\,(\#[]\mathbf{a})}{a\#\lambda[a]Y}\,(\#\mathsf{f})}{a\#X(\lambda[a]Y)}\,(\#\mathsf{f})$$

The following are *non*-derivable freshnesses in this signature:

$$\nvdash a\#a \qquad \nvdash a\#X(\lambda[a]Y) \qquad \nvdash a\#(\lambda[a]b)a$$

In the signature of theory FOL (Example 2.17), derivable freshnesses are:

$$\vdash a\#\forall[a]P \qquad a\#T\vdash a\#(a\approx a)[a\mapsto T] \qquad a\#X\vdash b\#(b\ a)\cdot X.$$

Non-derivable freshnesses in this signature are:

$$\nvdash a\#\forall[b]P \qquad \nvdash a\#(a\approx a)[a\mapsto T] \qquad a\#X\nvdash a\#(b\ a)\cdot X.$$

In order to define derivability on equalities recall notation $\pi\cdot t$, $t\sigma$ and t^π for actions on terms t from Definitions 3.1, 3.4 and 3.7. Also recall notation $\Delta\sigma$ and Δ^π for the extensions to freshness contexts Δ from Definition 3.10. Finally, recall from Definition 2.12 that as well as Δ we write ∇ for freshness contexts.

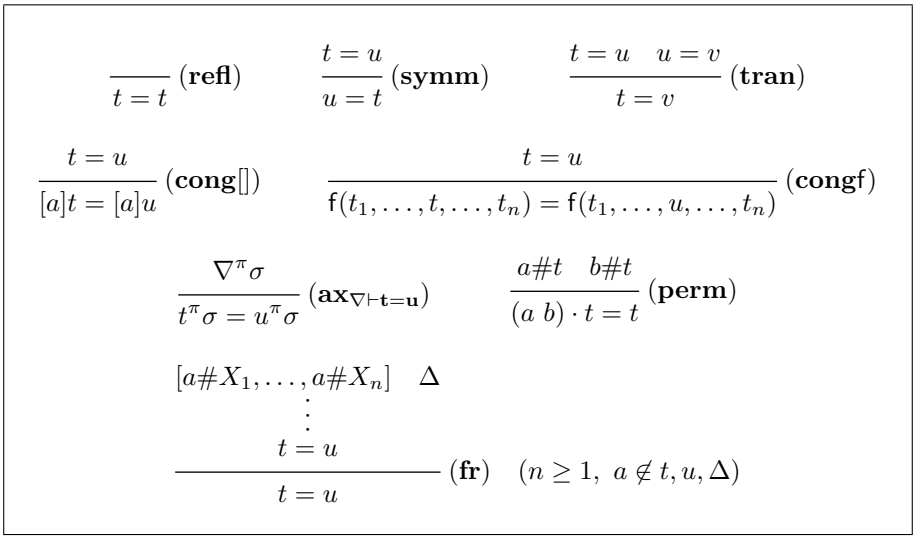


Figure 2: Derivation rules for equality

Definition 3.13. Define **derivability on equalities** (between terms in the signature of T) by the rules in Fig. 2. Write $\Delta \vdash_{\mathsf{T}} t = u$ when a derivation of $t = u$ exists using the rules in Figure 2 (and the ones in Figure 1), such that:

- for each instance of ($\mathbf{ax}_{\nabla \vdash \mathbf{t=u}}$), $\nabla \vdash t = u$ is an axiom of T;
- in the derivations of freshnesses (introduced by instances of ($\mathbf{ax}_{\nabla \vdash \mathbf{t=u}}$) and (**perm**)) the assumptions used are from Δ only.

Say that $t = u$ **is derivable from** Δ **in** T.

Write $\Delta \nvdash_{\mathsf{T}} t = u$ when there does not exist such a derivation.

The (**fr**) rule allows us to introduce freshness assumptions $a\#X_1, \ldots, a\#X_n$ into the derivation of $t = u$ from assumptions Δ, for atoms a that do not occur at all in t, u or Δ. The square brackets denote *discharge* in the sense of natural deduction (as in implication introduction) of these extra assumptions $a\#X_1, \ldots, a\#X_n$.

The rules (**refl**), (**symm**) and (**tran**) ensure that equality is an equivalence relation, and rules (**cong**[]) and (**congf**) ensure that it is a congruence. The rules ($\mathbf{ax}_{\nabla \vdash \mathbf{t=u}}$), (**perm**) and (**fr**) are discussed in more detail below.

Examples of derivability of equality can be found in the rest of this section.

3.2.1 The ($\mathbf{ax}_{\nabla \vdash \mathbf{t=u}}$) rule: instantiating axioms

($\mathbf{ax}_{\nabla \vdash \mathbf{t=u}}$) allows us to permutatively rename atoms and to instantiate unknowns. This gives the effect that atoms in axioms can be understood to range over *any*

(distinct) atoms, and unknowns can be understood to range over *any* terms. So

$$\frac{}{(\lambda[b]a)b = a[b \mapsto b]}\,(\mathbf{ax}_\beta) \qquad \frac{}{(\lambda[b]b)a = b[b \mapsto a]}\,(\mathbf{ax}_\beta)$$

are valid derivations in theory LAM (Example 2.17). The right derivation shows that substitution does not avoid capture, reflecting informal practice.

The use of the $(\mathbf{ax}_{\nabla \vdash \mathbf{t}=\mathbf{u}})$ rule introduces new proof obligations on the freshness side-conditions in ∇, as the following derivations show:

$$\frac{\dfrac{}{a\#b}\,(\#\mathbf{ab})}{\lambda[a](ba) = b}\,(\mathbf{ax}_\eta) \qquad \frac{a\#a}{\lambda[a](aa) = a}\,(\mathbf{ax}_\eta)$$

The left derivation is valid but the right one is not, because $a\#a$ is not derivable.

Note that instantiation of axioms $(\mathbf{abs}{\mapsto})$ and $(\mathbf{ren}{\mapsto})$ from theory SUB (Example 2.17), which both mention distinct atoms a and b, can never identify these atoms. For example,

$$\frac{c\#X}{([c]Y)[c \mapsto X] = [c](Y[c \mapsto X])}\,(\mathbf{ax}_{\mathbf{abs}\mapsto})$$

is not a valid instance of $(\mathbf{abs}{\mapsto})$ (even when $c\#X$ is derivable) since permutations are bijective: there is no π such that both $\pi(a) = c$ and $\pi(b) = c$.

In informal practice, derivations are often presented in a calculational style, e.g. the sequence of equalities

$$\lambda x.(((\lambda x.y)x)x) =_\beta \lambda x.(yx) =_\eta y$$

represents that two terms can be related by (reading from left to right) first applying β-conversion followed by η-conversion. This is fully formally represented by the following derivation:

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{(\lambda[a]b)a = b[a \mapsto a]}\,(\mathbf{ax}_\beta) \quad \dfrac{\dfrac{}{a\#b}\,(\#\mathbf{ab})}{b[a \mapsto a] = b}\,(\mathbf{ax}_{\#\mapsto})}{(\lambda[a]b)a = b}\,(\mathbf{tran})}{((\lambda[a]b)a)a = ba}\,(\mathbf{congf})}{[a](((\lambda[a]b)a)a) = [a](ba)}\,(\mathbf{cong}[])}{\lambda[a](((\lambda[a]b)a)a) = \lambda[a](ba)}\,(\mathbf{congf}) \quad \dfrac{\dfrac{}{a\#b}\,(\#\mathbf{ab})}{\lambda[a](ba) = b}\,(\mathbf{ax}_\eta)}{\lambda[a](((\lambda[a]b)a)a) = b}\,(\mathbf{tran})$$

Reading the derivation bottom-up, the instance of (**tran**) in the conclusion introduces two equalities that correspond directly to the two equations in the above

sequence of equalities. The derivation of the right equality takes care of η-equality: it instantiates the (η) axiom, and shows that the freshness side-condition is satisfied. The derivation of the left equality takes care of the β-equality: it instantiates the (β) axiom, which introduces a substitution, and we show how this substitution is applied.

Remark 3.14. Another version of the $(\mathbf{ax}_{\nabla\vdash\mathbf{t}=\mathbf{u}})$ rule is possible, which uses the object-level action $\pi \cdot t$ instead of the meta-level action t^π:

$$\frac{\pi \cdot \nabla\sigma}{\pi \cdot t\sigma = \pi \cdot u\sigma}\,(\mathbf{ax}'_{\nabla\vdash\mathbf{t}=\mathbf{u}}).$$

However in this case, atoms in substitution σ are renamed according to π. For example, from the axiom $[a]X = [b]X$ it is immediate that $[b]a = [a]a$ is derivable with $(\mathbf{ax}_{\mathbf{[a]X=[b]X}})$ where we choose $\pi = (b\ a)$ and $\sigma = [a/X]$. It is also derivable with $(\mathbf{ax}'_{\mathbf{[a]X=[b]X}})$ but we must choose $\pi = (b\ a)$ and $\sigma = [b/X]$. We find this version less natural.

3.2.2 The (perm) rule: α-equivalence

The (**perm**) rule provides us with a concise way of expressing α-equivalence. To illustrate this, the following derivations are valid in CORE (the theory with no axioms):

$$\dfrac{\dfrac{\dfrac{}{a\#b}\,(\#\mathbf{ab})}{a\#[b]b}\,(\#[]\mathbf{b}) \qquad \dfrac{}{b\#[b]b}\,(\#[]\mathbf{a})}{[a]a = [b]b}\,(\mathbf{perm}) \qquad\qquad \dfrac{\dfrac{a\#X}{a\#[b]X}\,(\#[]\mathbf{b}) \qquad \dfrac{}{b\#[b]X}\,(\#[]\mathbf{a})}{[a](b\ a)\cdot X = [b]X}\,(\mathbf{perm})$$

So $\vdash_{\mathsf{CORE}} [a]a = [b]b$ and $a\#X \vdash_{\mathsf{CORE}} [a](b\ a)\cdot X = [b]X$. To see that the instances of (**perm**) are valid, we note that $[a]a \equiv (b\ a)\cdot[b]b$ and $[a](b\ a)\cdot X \equiv (b\ a)\cdot[b]X$.

As another example, we show how we use (**perm**) rule to rename a bound variable in a λ-term. Consider the following derivation in the λ-calculus:

$$(\lambda x.xx)(\lambda x.\lambda y.xy) =_\beta (\lambda x.\lambda y.xy)(\lambda x.\lambda y.xy) =_\beta \lambda y.(\lambda x.\lambda y.xy)y =_\beta \lambda y.\lambda z.yz.$$

In the last step the bound variable x is implicitly renamed during β-reduction to avoid capture. Nominal algebra makes this explicit. We present the nominal algebra derivation of this last step as a calculation:

$$\begin{array}{ll} & \lambda[b]\underline{(\lambda[a]\lambda[b]ab)b} \\ = & \quad \{\ \text{axiom } (\beta)\ \} \\ & \lambda[b]\underline{(\lambda[b]ab)[a \mapsto b]} \\ = & \quad \{\ \text{axiom } (\mathsf{f}{\mapsto})\ \} \end{array}$$

$$
\begin{array}{ll}
 & \lambda[b]\lambda(\underline{[b]ab})[a \mapsto b] \\
= & \quad \{\ (\mathbf{perm}), \text{ since } \vdash b\#[c]ac \text{ and } \vdash c\#[c]ac\ \} \\
 & \lambda[b]\lambda(\underline{[c]ac})[a \mapsto b] \\
= & \quad \{\ \text{axiom } (\mathbf{abs}{\mapsto}), \text{ since } \vdash c\#b\ \} \\
 & \lambda[b]\lambda[c]\underline{(ac)[a \mapsto b]} \\
= & \quad \{\ \text{axiom } (\mathsf{f}{\mapsto})\ \} \\
 & \lambda[b]\lambda[c](\underline{a[a \mapsto b]})(c[a \mapsto b]) \\
= & \quad \{\ \text{axiom } (\mathbf{var}{\mapsto})\ \} \\
 & \lambda[b]\lambda[c]b(\underline{c[a \mapsto b]}) \\
= & \quad \{\ \text{axiom } (\#{\mapsto}), \text{ since } \vdash a\#c\ \} \\
 & \lambda[b]\lambda[c]bc
\end{array}
$$

In each step of the calculation, we have indicated in the hint which derivation rule is applied and which freshness constraints it had to satisfy (if any), and we have underlined the subterm on which the axiom is applied. From this information the full derivation can be reconstructed using (**cong**[]), (**congf**) and (**tran**).

As a final example we show that we can rename an atom which is substituted for using explicit substitution term-former sub from theory SUB (Example 2.17):

Lemma 3.15. $b\#X \vdash_{\mathsf{CORE}} X[a \mapsto T] = ((b\ a) \cdot X)[b \mapsto T]$

Proof. De-sugaring, we derive $\mathsf{sub}([a]X, T) = \mathsf{sub}([b](b\ a) \cdot X, T)$ from $b\#X$:

$$
\dfrac{\dfrac{\dfrac{}{a\#[a]X}\,(\#[]\mathbf{a}) \qquad \dfrac{b\#X}{b\#[a]X}\,(\#[]\mathbf{b})}{[b](b\ a) \cdot X = [a]X}\,(\mathbf{perm})}{[a]X = [b](b\ a) \cdot X}\,(\mathbf{symm})}{\mathsf{sub}([a]X, T) = \mathsf{sub}([b](b\ a) \cdot X, T)}\,(\mathbf{congf})
$$

□

In Subsection 4.4 we will show that derivability in CORE precisely corresponds to α-equivalence on nominal terms, in the sense of nominal unification [UPG04, Figure 2] and nominal rewriting [FG07, p.13].

Remark 3.16. (**perm**) admits a representation as an axiom:

$$a\#X, b\#X \vdash (a\ b) \cdot X = X.$$

It is not hard to see that with this axiom, (**perm**) becomes an instance of (**ax**). Our intended semantics is in nominal sets so (**perm**) is mandatory for soundness (Theorem 6.7) to hold, and accordingly we build it into the derivation system.

3.2.3 The (fr) rule: introducing fresh atoms

The (**fr**) rule allows us to introduce a fresh atom into the derivation. It mirrors the generation of a fresh name in rules such as the $\forall$ right-introduction rule 'from $\Gamma \vdash \phi$ derive $\Gamma \vdash \forall x.\phi$ provided x is not free in Γ'. In a sequent style presentation of nominal algebra, (**fr**) would be

$$\frac{\Delta, a\#X_1, \ldots, a\#X_n \vdash t = u}{\Delta \vdash t = u} \quad (n \geq 1, a \notin t, u, \Delta).$$

To illustrate the extra power (**fr**) gives us, we consider a theory C with one axiom $a\#X \vdash X = a$.

Lemma 3.17. *We can derive* $\vdash_{\mathsf{C}} X = Y$ with (**fr**)*, but we cannot* without *it.*

Proof. The following is a derivation of $\vdash_{\mathsf{C}} X = Y$ with (**fr**):

$$\dfrac{\dfrac{\dfrac{[a\#X]^1}{X = a}\,(\mathbf{ax_{a\#X\vdash X=a}}) \qquad \dfrac{\dfrac{[a\#Y]^1}{Y = a}\,(\mathbf{ax_{a\#X\vdash X=a}})}{a = Y}\,(\mathbf{symm})}{X = Y}\,(\mathbf{tran})}{X = Y}\,(\mathbf{fr})^1$$

In the above derivation, the superscript number one ^{1} is an annotation associating the instance of the rule (**fr**) with the assumptions it discharges in the derivation.

In order to show that it is impossible to derive $\vdash_{\mathsf{C}} X = Y$ without (**fr**), we show the more general property that we can never conclude equations of the form $X = t$ or $t = X$, where $t \not\equiv X$, without the use of (**fr**).

We proceed by contradiction. Suppose we *can* conclude equations of the form $X = t$ and $t = X$ where $t \not\equiv X$. Looking at the structure of the derivation rules, it is easy to see that the (**refl**), (**cong**[]) and (**congf**) rules could not have been applied. Also we can see that we could never have applied the ($\mathbf{ax_{a\#X\vdash X=a}}$) and (**perm**) rules: the derivations of the freshness side-conditions belonging to these rules require the presence of at least one freshness assumption, but this is impossible since we work in the empty freshness context. So the only rules that could have been applied are (**symm**) and (**tran**).

We only consider the $X = t$ case, the $t = X$ case is completely analogous. Suppose the derivation of $X = t$ concludes in:

- (**symm**). Then $X = t$ is derived from $t = X$, which fits the pattern.
- (**tran**). Then $X = t$ is derived from $X = u$ and $u = t$. Now there are two cases: either $u \not\equiv X$, in which case the first equation fits the pattern, or $u \equiv X$, in which case the second equation fits the pattern.

So using (**symm**) and (**tran**) we cannot conclude the derivation. We have arrived at a contradiction and the result follows. □

Another example that shows the extra power of the (**fr**) rule is derivability of $X[a \mapsto a] = X$ in theory SUB (Example 2.17).

Lemma 3.18. $\vdash_{\mathsf{SUB}} X[a \mapsto a] = X$.

Proof. We derive $X[a \mapsto a] = X$ as follows:

$$\dfrac{\dfrac{\dfrac{\dfrac{}{a\#[a]X}\,(\#[]\mathbf{a}) \qquad \dfrac{[b\#X]^1}{b\#[a]X}\,(\#[]\mathbf{b})}{[b](b\ a)\cdot X = [a]X}\,(\mathbf{perm})}{[a]X = [b](b\ a)\cdot X}\,(\mathbf{symm})}{X[a \mapsto a] = ((b\ a)\cdot X)[b \mapsto a]}\,(\mathbf{congf}) \qquad \dfrac{\dfrac{[b\#X]^1}{a\#(b\ a)\cdot X}\,(\#\mathbf{X})}{((b\ a)\cdot X)[b \mapsto a] = X}\,(\mathbf{ax}_{\mathbf{ren}\mapsto})}{\dfrac{X[a \mapsto a] = X}{X[a \mapsto a] = X}\,(\mathbf{fr})^1}\,(\mathbf{tran})$$

The instance of the (**ren**$\mapsto$) axiom is valid since we have used the fact that $X \equiv (a\ b)\cdot(b\ a)\cdot X$ in the right-hand side of the equation. □

We conjecture that we cannot derive $X[a \mapsto a] = X$ in SUB *without* (**fr**). Intuitively this is because to α-rename a so that we can use (**ren**$\mapsto$), we need an atom fresh for X.

Remark 3.19. The (**fr**) rule can be simulated by a number of steps using the following more compact rule:

$$\dfrac{\begin{array}{c}[a\#X] \quad \Delta \\ \vdots \\ t = u\end{array}}{t = u}\,(\mathbf{fr}') \quad (a \notin t, u)$$

Because of its simplicity, this rule is easier to use in inductive proofs on the derivation rules for equality. However, we prefer to use (**fr**) in this paper, because it corresponds more closely to the informal notion of 'let a be a fresh atom'.

4 Proof-Theoretical Results

We provide a number of proof-theoretical results for freshness and equality that will be used throughout this paper.

4.1 Freshness

Lemma 4.1. *It is decidable whether* $\Delta \vdash a\#t$, *for any* Δ, a *and* t.

Proof. We observe that the derivation rules of Figure 1 are syntax-directed. □

The rules for freshness also hold in the opposite direction:

Lemma 4.2.

1. *If* $\Delta \vdash a\#X$ *then* $a\#X \in \Delta$.
2. *If* $\Delta \vdash a\#\pi \cdot X$ *then* $\Delta \vdash \pi^{-1}(a)\#X$.
3. *If* $\Delta \vdash a\#[b]t$ *then* $\Delta \vdash a\#t$.
4. *If* $\Delta \vdash a\#\mathsf{f}(t_1, \ldots, t_n)$ *then* $\Delta \vdash a\#t_i$, *for all* i, $1 \leq i \leq n$.

Proof. By an induction on the structure of the derivation rules in Figure 1. □

4.2 Equivariance

Definition 4.3. We naturally extend notation for t^π and Δ^π to theories: given a theory $\mathsf{T} = (\Sigma, Ax)$, write T^π for (Σ, Ax^π) such that $\nabla^\pi \vdash t^\pi = u^\pi \in Ax^\pi$ if and only if $\nabla \vdash t = u \in Ax$.

Lemma 4.4. *If* $\Delta \vdash_{\mathsf{T}} t = u$ *then* $\Delta \vdash_{\mathsf{T}^\pi} t = u$.

Proof. By induction on derivations. The only non-trivial case is $(\mathbf{ax}_{\nabla \vdash \mathbf{t}=\mathbf{u}})$, where after applying the inductive hypothesis we need to show that

$$\Delta \vdash_{\mathsf{T}^\pi} \nabla^{\pi'}\sigma \quad \text{implies} \quad \Delta \vdash_{\mathsf{T}^\pi} t^{\pi'}\sigma = u^{\pi'}\sigma.$$

By Lemma 3.8, it is equivalent to show that

$$\Delta \vdash_{\mathsf{T}^\pi} \nabla^{\pi^{\pi'\circ\pi^{-1}}}\sigma \quad \text{implies} \quad \Delta \vdash_{\mathsf{T}^\pi} t^{\pi^{\pi'\circ\pi^{-1}}}\sigma = u^{\pi^{\pi'\circ\pi^{-1}}}\sigma.$$

This follows by $(\mathbf{ax}_{\nabla^\pi \vdash \mathbf{t}^\pi=\mathbf{u}^\pi})$ taking permutation $\pi' \circ \pi^{-1}$ and substitution σ. □

Theorem 4.5 states that we may permute atoms in freshnesses and equality at the *meta-level*. This property is one way in which nominal algebra reflects internally the properties of atoms, as expressed by Theorem A.4, which give 'nominal techniques' much of their character:

Theorem 4.5 (Meta-level equivariance)**.** *For any* π*:*

1. *if* $\Delta \vdash a\#t$ *then* $\Delta^\pi \vdash \pi(a)\#t^\pi$*;*
2. *if* $\Delta \vdash_{\mathsf{T}} t = u$ *then* $\Delta^\pi \vdash_{\mathsf{T}} t^\pi = u^\pi$.

Proof. For the second case suppose that $\Delta \vdash_{\mathsf{T}} t = u$. By equivariance (Theorem A.4) also $\Delta^\pi \vdash_{\mathsf{T}^\pi} t^\pi = u^\pi$.[3] Then by Lemma 4.4 we obtain

$$\Delta^\pi \vdash_{\mathsf{T}^{\pi\pi^{-1}}} t^\pi = u^\pi.$$

Using Lemma 3.8 we can easily show that $\mathsf{T}^{\pi^{\pi^{-1}}}$ is syntactically equivalent to T, so we obtain $\Delta^\pi \vdash_{\mathsf{T}} t^\pi = u^\pi$ as required.

The proof of the first case is simpler, since it does not refer to theory T. □

[3] Intuitively: however we may have established that $\Delta \vdash_{\mathsf{T}} t = u$, since atoms have no internal structure the same method will work for any permutation of those atoms.

We can also permute atoms in freshnesses and equations at the *object-level* (without changing the freshness context):

Theorem 4.6 (Object-level equivariance)**.** *For any π:*

1. *if $\Delta \vdash a\#t$ then $\Delta \vdash \pi(a)\#\pi \cdot t$;*
2. *if $\Delta \vdash_{\mathsf{T}} t = u$ then $\Delta \vdash_{\mathsf{T}} \pi \cdot t = \pi \cdot u$.*

Proof. By induction on the structure of derivations. We consider the most interesting cases only. Suppose the derivation concludes in...

- ($\#\mathbf{X}$). Then $a\#\pi' \cdot X$ is derived from ${\pi'}^{-1}(a)\#X$, for some $\pi' \neq \iota$, and we need to show $\pi(a)\#\pi \cdot (\pi' \cdot X)$. By Lemma 3.2, this is equivalent to $\pi(a)\#(\pi \circ \pi') \cdot X$. We continue by case distinction:
  - If $\pi \circ \pi' = \iota$ then $\pi(a)\#(\pi \circ \pi') \cdot X$ is equivalent to the assumption ${\pi'}^{-1}(a)\#X$, since $\pi = {\pi'}^{-1}$ by basic permutation group theory.
  - If $\pi \circ \pi' \neq \iota$ then $\pi(a)\#(\pi \circ \pi') \cdot X$ follows from $(\pi \circ \pi')^{-1}(\pi(a))\#X$ by ($\#\mathbf{X}$) (which may now be applied). This is equivalent to the assumption ${\pi'}^{-1}(a)\#X$, since $(\pi \circ \pi')^{-1}(\pi(a)) = {\pi'}^{-1}(a)$.
- ($\mathbf{ax}_{\nabla \vdash \mathbf{t=u}}$). Then $t^{\pi'}\sigma = u^{\pi'}\sigma$ is derived and $\Delta \vdash_{\mathsf{T}} \nabla^{\pi'}\sigma$. We need to derive $\pi \cdot t^{\pi'}\sigma = \pi \cdot u^{\pi'}\sigma$ from Δ. By Lemma 3.6, $\pi \cdot t^{\pi'}\sigma = \pi \cdot u^{\pi'}\sigma$ is equivalent to $(\pi \cdot t^{\pi'})\sigma = (\pi \cdot u^{\pi'})\sigma$. Now let σ' map each $X \in \nabla, t, u$ to $\pi \cdot X$, then by Lemma 3.9 it suffices to derive

  $$t^{\pi'\pi}(\sigma' \circ \sigma) = u^{\pi'\pi}(\sigma' \circ \sigma).$$

  By Lemma 3.8, this is equivalent to $t^{\pi \circ \pi'}(\sigma' \circ \sigma) = u^{\pi \circ \pi'}(\sigma' \circ \sigma)$. Now this follows from $\nabla^{\pi \circ \pi'}(\sigma' \circ \sigma)$ by ($\mathbf{ax}_{\nabla \vdash \mathbf{t=u}}$) with permutation $\pi \circ \pi'$ and substitution $\sigma' \circ \sigma$. By Lemmas 3.6, 3.9 and 3.8 this is equivalent to $\pi \cdot \nabla^{\pi'}\sigma$. We are done since this follows from Δ by the inductive hypothesis.
- ($\mathbf{fr}$). Then $\Delta, a\#X_1, \ldots, a\#X_n \vdash_{\mathsf{T}} t = u$ for some $a \notin \Delta, t, u$ and we assume the inductive hypothesis of this derivation. If $\pi(a) = a$ there is no problem since then $a \notin \Delta, \pi \cdot t, \pi \cdot u$ and we may extend the derivation with ($\mathbf{fr}$).
  However, suppose $\pi(a) \neq a$ and so (possibly) $a \in \pi \cdot t, \pi \cdot u$. We observe that the predicate

  > "if the labelled tree Π is a valid derivation of $\Delta \vdash_{\mathsf{T}} t = u$, then for all permutations π' there are derivations of $\Delta \vdash_{\mathsf{T}} \pi' \cdot t = \pi' \cdot u$"

  has free variables Π, Δ, T, t and u.
  By equivariance (Theorem A.4), the precidate above holds of $\Pi^{(a'\ a)}$, $\Delta^{(a'\ a)}$, $\mathsf{T}^{(a'\ a)}$, $t^{(a'\ a)}$ and $u^{(a'\ a)}$ (the informal notation $\Pi^{(a'\ a)}$ denotes Π in which all atoms are permuted according to $(a'\ a)$). Now using Lemma 4.4 we deduce the inductive hypothesis of $\Delta, a'\#X_1, \ldots, a'\#X_n \vdash_{\mathsf{T}} t = u$ for any $a' \notin \Delta, t, u, \pi$. Then $\Delta, a'\#X_1, \ldots, a'\#X_n \vdash_{\mathsf{T}} \pi \cdot t = \pi \cdot u$ and we extend the derivation with ($\mathbf{fr}$) to deduce $\Delta \vdash_{\mathsf{T}} \pi \cdot t = \pi \cdot u$ as required.

□

Note that instead of using ZFA equivariance in the (**fr**) case of the above proof, we could have proven the theorem by an induction on the *depth* of derivations. We prefer equivariance however, because it allows for a more compact and readable proof.

As a corollary we obtain the following congruence property.

Corollary 4.7 (Congruence)**.** *For any X, v:*

$$\textit{If} \quad \Delta \vdash_{\mathsf{T}} t = u \quad \textit{then} \quad \Delta \vdash_{\mathsf{T}} v[t/X] = v[u/X].$$

Proof. By an induction on the structure of v. For the case of $v \equiv \pi \cdot X$ we use Theorem 4.6. □

4.3 Substitution, weakening and strengthening

We can substitute terms for unknowns provided those terms violate no freshness assumptions made on the unknowns:

Theorem 4.8 (Meta-level substitution)**.** *For any Δ', Δ, σ such that $\Delta' \vdash \Delta\sigma$:*

1. *if $\Delta \vdash a\#t$ then $\Delta' \vdash a\#t\sigma$;*
2. *if $\Delta \vdash_{\mathsf{T}} t = u$ then $\Delta' \vdash_{\mathsf{T}} t\sigma = u\sigma$.*

Proof. Natural deduction derivations are such that the conclusion of one derivation may be 'plugged in' to an assumption in another derivation. For (#**X**) we use object-level equivariance (Theorem 4.6). For (**fr**) we might have to use equivariance (Theorem A.4) to rename the freshly chosen atom a if it is mentioned by Δ', $t\sigma$ or $u\sigma$. □

The above condition $\Delta' \vdash \Delta\sigma$ ensures that $\Delta\sigma$ is *consistent*, in the sense that $a\#\sigma(X)$ is derivable for each $a\#X \in \Delta$.

Corollary 4.9. *For closed t, u, if $\vdash_{\mathsf{T}} t = u$ then it has a derivation that does not mention any unknowns or instances of* (#**X**) *or* (**fr**)*.*

Proof. Let Π be a derivation of $\vdash_{\mathsf{T}} t = u$. Now take c to be an atom that does not occur anywhere in Π, and let Π' be Π in which:

- each unknown X is mapped to c;
- each instance of (#**X**) is replaced by (#**ab**); that is, each instance of (#**X**) is of the form
$$\frac{[\pi^{-1}(a)\#X]}{a\#\pi \cdot X}\,(\#\mathbf{X}),$$
where square brackets denote discharge of the freshness assumption. This is replaced by
$$\frac{}{a\#c}\,(\#\mathbf{ab}).$$

Note that we write c instead of $\pi(c)$ since $\pi(c) = c$.

Using Theorem 4.8 it is not hard to see that Π' is a valid derivation of $t = u$ that does not mention unknowns. Since the instances of (**fr**) do not discharge any assumptions anymore (they have been removed in the previous step), these instances be safely removed. The result follows. □

Another corollary of Theorem 4.8 is that we can *weaken* the freshness context in derivations:

Corollary 4.10 (Weakening)**.** *If $\Delta \subseteq \Delta'$ then:*

1. *if $\Delta \vdash a\#t$ then $\Delta' \vdash a\#t$;*
2. *if $\Delta \vdash_{\mathsf{T}} t = u$ then $\Delta' \vdash_{\mathsf{T}} t = u$.*

Proof. By meta-level substitution (Theorem 4.8), using the empty substitution $\sigma = []$, the proof obligations follow from $\Delta' \vdash \Delta$. So for each $a\#X \in \Delta$, we must show $\Delta' \vdash a\#X$. This is trivial since $\Delta \subseteq \Delta'$. □

The (**fr**) rule already stated when a freshness context may be *strengthened* on equalities. This also holds for freshnesses:

Lemma 4.11 (Strengthening)**.**

1. *If $\Delta, a\#X_1, \ldots, a\#X_n \vdash b\#t$, where $n \geq 1$ and $a \notin \Delta, t$, then $\Delta \vdash b\#t$.*
2. *If $\Delta, a\#X_1, \ldots, a\#X_n \vdash_{\mathsf{T}} t = u$, where $n \geq 1$ and $a \notin \Delta, t, u$, then $\Delta \vdash_{\mathsf{T}} t = u$.*

Proof. The equational case is precisely (**fr**). For the freshness case, we inductively transform a derivation of $\Delta, a\#X_1, \ldots, a\#X_n \vdash b\#t$ to a derivation of $\Delta \vdash b\#t$:

- If $\Delta, a\#X_1, \ldots, a\#X_n \vdash b\#X$ by assumption, then $b\#X \in \Delta$, so $\Delta \vdash b\#X$ by assumption.
- (#**ab**) and (#[]**a**) carry over directly.
- (#**X**), (#[]**b**) and (#f) follow using the inductive hypothesis and the following properties: if $a \notin \pi \cdot X$ then $a \notin X$, if $a \notin [c]t$ then $a \notin t$, and if $a \notin \mathsf{f}(t_1, \ldots, t_n)$ then $a \notin t_i$ for all i.

□

4.4 α-Equivalence with Meta-Variables

In Example 2.17 we defined CORE as a family of nominal algebra theories with no axioms (one for each signature). In Subsection 3.2 we have given some examples to show that the (**perm**) rule expresses α-equivalence with meta-variables.

In this section we will prove that theory CORE precisely corresponds to the existing syntax-directed notion of α-equivalence on nominal terms from [UPG04,

FG07]. We will use this correspondence to show that CORE-equality is decidable, and that theory CORE is *consistent* (does not equate *all* terms.

Definition 4.14 was introduced in [UPG04, Figure 2]; the proofs are modelled on a method presented in [FG07, p.13].

Definition 4.12. We write $\mathrm{ds}(\pi, \pi')$ for the **difference set** $\{a \mid \pi(a) \neq \pi'(a)\}$ of π and π'. We write $\mathrm{ds}(\pi, \pi')\#t$ for the set of freshnesses $\{a\#t \mid \pi(a) \neq \pi'(a)\}$.

Lemma 4.13. *If* $\Delta \vdash \mathrm{ds}(\pi, \pi')\#t$ *then* $\Delta \vdash_{\mathsf{CORE}} \pi \cdot t = \pi' \cdot t$.

Proof. We work by induction on the number of elements in $\mathrm{ds}(\pi, \pi')$. If this set is empty then $\pi = \pi'$ and the result follows easily by (**refl**). Now suppose $a \in \mathrm{ds}(\pi, \pi')$. We construct a partial derivation of the proof obligation:

$$\dfrac{\pi \cdot t = ((\pi(a)\ \pi'(a)) \circ \pi') \cdot t \qquad \dfrac{\pi(a)\#\pi' \cdot t \quad \pi'(a)\#\pi' \cdot t}{((\pi(a)\ \pi'(a)) \circ \pi') \cdot t = \pi' \cdot t}\,(\mathbf{perm})}{\pi \cdot t = \pi' \cdot t}\,(\mathbf{tran})$$

The following proof obligations remain:

1. $\pi \cdot t = ((\pi(a)\ \pi'(a)) \circ \pi') \cdot t$ follows from $\mathrm{ds}(\pi, (\pi(a)\ \pi'(a)) \circ \pi')\#t$ by the inductive hypothesis, provided $|\mathrm{ds}(\pi, (\pi(a)\ \pi'(a)) \circ \pi')| < |\mathrm{ds}(\pi, \pi')|$. This condition is satisfied, since $\mathrm{ds}(\pi, (\pi(a)\ \pi'(a)) \circ \pi') = \mathrm{ds}(\pi, \pi') \setminus \{a\}$. Finally, the remaining proof obligation $\mathrm{ds}(\pi, (\pi(a)\ \pi'(a)) \circ \pi')\#t$ follows from assumption $\mathrm{ds}(\pi, \pi')\#t$.
2. $\pi(a)\#\pi' \cdot t$ follows from $\pi'^{-1}(\pi(a))\#t$ by object-level equivariance (Theorem 4.6). Now this is one of the assumptions $\mathrm{ds}(\pi, \pi')\#t$: by Definition 4.12, $\pi'^{-1}(\pi(a)) \in \mathrm{ds}(\pi, \pi')$ when $\pi(\pi'^{-1}((\pi(a))) \neq \pi(a)$, and, using the fact that $\neq$ is invariant under permutation, this follows from the assumption $\pi(a) \neq \pi'(a)$.
3. $\pi'(a)\#\pi' \cdot t$ follows from $a\#t$ by object-level equivariance (Theorem 4.6). This follows directly from assumption $\mathrm{ds}(\pi, \pi')\#t$, since $a \in \mathrm{ds}(\pi, \pi')$.

□

Definition 4.14. Let $t \approx_\Delta u$ be an ordered tuple of a term t, a freshness context Δ, and a term u. Let the **derivable equalities of** $t \approx_\Delta u$ be inductively defined by the rules in Figure 3.

Theorem 4.15 (Equivalence of CORE and $\approx_\Delta$)**.** $\Delta \vdash_{\mathsf{CORE}} t = u$ *if and only if* $t \approx_\Delta u$ *is derivable using the rules of Figure 3.*

Proof. The left-to-right direction is by induction on the structure of nominal algebra derivations of $\Delta \vdash_{\mathsf{CORE}} t = u$. By the inductive hypothesis it suffices to show:

- Syntax-directed equality $\approx_\Delta$ is an equivalence relation and a congruence. This is [FG07, Theorem 24].

$$\frac{}{a \approx_\Delta a}\,(\mathbf{Ax}) \qquad \frac{\Delta \vdash \mathrm{ds}(\pi', \pi)\#X}{\pi \cdot X \approx_\Delta \pi' \cdot X}\,(\mathbf{Ds})$$

$$\frac{t \approx_\Delta u}{[a]t \approx_\Delta [a]u}\,(\mathbf{Absaa}) \qquad \frac{(b\ a) \cdot t \approx_\Delta u \quad \Delta \vdash b\#t}{[a]t \approx_\Delta [b]u}\,(\mathbf{Absab})$$

$$\frac{t_1 \approx_\Delta u_1 \quad \cdots \quad t_n \approx_\Delta u_n}{\mathsf{f}(t_1, \ldots, t_n) \approx_\Delta \mathsf{f}(u_1, \ldots, u_n)}\,(\mathbf{F})$$

Figure 3: Syntax-directed rules for CORE

- If $\Delta \vdash a\#t$ and $\Delta \vdash b\#t$ then $(a\ b) \cdot t \approx_\Delta t$. By induction on t.
- If $t \approx_{\Delta, a\#X_1, \ldots, a\#X_n} u$ where $a \notin t, u, \Delta$ then $t \approx_\Delta u$. By straightforward induction on the structure of derivations of $t \approx_{\Delta, a\#X_1, \ldots, a\#X_n} u$. The case of (**Absab**) uses strengthening (Lemma 4.11) to strengthen the assumption $\Delta, a\#X_1, \ldots, a\#X_n \vdash c\#t$ to $\Delta \vdash c\#t$.

For the right-to-left direction we work by induction on derivations of $t \approx_\Delta u$. By the inductive hypothesis it suffices to show:

- $\Delta \vdash_{\mathsf{CORE}} a = a$. This is an instance of (**refl**).
- If $\Delta \vdash \mathrm{ds}(\pi, \pi')\#X$ then $\Delta \vdash_{\mathsf{CORE}} \pi \cdot X = \pi' \cdot X$. By Lemma 4.13.
- If $\Delta \vdash_{\mathsf{CORE}} t_i = u_i$ for $1 \le i \le n$, then $\Delta \vdash_{\mathsf{CORE}} \mathsf{f}(t_1, \ldots, t_n) = \mathsf{f}(u_1, \ldots, u_n)$. Using a number of instances of (**tran**) and (**congf**).
- If $\Delta \vdash_{\mathsf{CORE}} t = u$ then $\Delta \vdash_{\mathsf{CORE}} [a]t = [a]u$. This is (**cong**[]).
- If $\Delta \vdash_{\mathsf{CORE}} (b\ a) \cdot t = u$ and $\Delta \vdash b\#t$ then $\Delta \vdash_{\mathsf{CORE}} [a]t = [b]u$. Suppose that Π and Π' are derivations of $\Delta \vdash_{\mathsf{CORE}} (b\ a) \cdot t = u$ and $\Delta \vdash b\#t$ respectively. Then the following is a derivation of $\Delta \vdash_{\mathsf{CORE}} [a]t = [b]u$:

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\vdots\ \Pi'}{b\#t}}{b\#[a]t}\,(\#[]\mathbf{b}) \qquad \dfrac{}{a\#[a]t}\,(\#[]\mathbf{a})}{[b](b\ a) \cdot t = [a]t}\,(\mathbf{perm})}{[a]t = [b](b\ a) \cdot t}\,(\mathbf{symm}) \qquad \dfrac{\vdots\ \Pi}{(b\ a) \cdot t = u}}{[b](b\ a) \cdot t = [b]u}\,(\mathbf{cong}[])}{[a]t = [b]u}\,(\mathbf{tran})$$

□

As corollaries of Theorem 4.15, we obtain syntactic criteria for determining equality in CORE, consistency of CORE, and preservation of freshness for CORE.

Corollary 4.16 (Decidability of CORE)**.** $\Delta \vdash_{\mathsf{CORE}} t = u$ *precisely when one of the following holds:*

1. $t \equiv a$ *and* $u \equiv a$.
2. $t \equiv \pi \cdot X$ *and* $u \equiv \pi' \cdot X$ *and* $\Delta \vdash \mathrm{ds}(\pi, \pi')\#X$.
3. $t \equiv [a]t'$ *and* $u \equiv [a]u'$ *and* $\Delta \vdash_{\mathsf{CORE}} t' = u'$.
4. $t \equiv [a]t'$ *and* $u \equiv [b]u'$ *and* $\Delta \vdash b\#t'$ *and* $\Delta \vdash_{\mathsf{CORE}} (b\ a) \cdot t' = u'$.
5. $t \equiv \mathsf{f}(t_1, \ldots, t_n)$ *and* $u \equiv \mathsf{f}(u_1, \ldots, u_n)$ *and* $\Delta \vdash_{\mathsf{CORE}} t_i = u_i$ *for* $1 \leq i \leq n$.

Proof. By Theorem 4.15 it suffices to inspect the rules for $t \approx_\Delta u$, which are just a rendering of the above criteria in terms of derivation rules. □

Corollary 4.17 (Consistency of CORE)**.** *For all* Δ *there are* t *and* u *such that* $\Delta \nvdash_{\mathsf{CORE}} t = u$.

Proof. By Corollary 4.16, $\Delta \vdash_{\mathsf{CORE}} a = b$ is never derivable. □

Corollary 4.18 (Preservation of freshness for CORE)**.** *If* $\Delta \vdash_{\mathsf{CORE}} t = u$ *then*

$$\Delta \vdash a\#t \quad \textit{if and only if} \quad \Delta \vdash a\#u.$$

Proof. By induction on t using the syntactic criteria of Corollary 4.16. □

5 Nominal Sets

In this section we give a brief overview of nominal sets. For full treatments we refer the reader to [GP02] or [Pit03] ([Pit03] contains a simplified presentation of [GP02]).

Recall from Definitions 2.3 and 2.4 that we write $\mathbb{A}$ for the set of all atoms, $\mathbb{P}$ for the set of all permutations, ι and $\circ$ for the identity and composition of permutations, and recall from Definition 4.12 that we write $\mathrm{ds}(\pi, \pi')$ for the difference set of π and π'.

Definition 5.1. A **$\mathbb{P}$-action** $\cdot$ on a set $\mathbb{X}$ is a function $\cdot : \mathbb{P} \times \mathbb{X} \to \mathbb{X}$, write it infix as $\pi \cdot x$, such that $\iota \cdot x = x$ and $\pi \cdot (\pi' \cdot x) = (\pi \circ \pi') \cdot x$ for all $x \in \mathbb{X}$. Say that a *finite* set of atoms A **supports** x when for any permutation π:

$$\text{if } \pi(a) = a \text{ for each } a \in A, \text{ then } \pi \cdot x = x.$$

Say that x has **finite support** when there exists such a set of atoms.

A **nominal set** is a set $\mathbb{X}$ equipped with a $\mathbb{P}$-action on $\mathbb{X}$ such that each $x \in \mathbb{X}$ has finite support.

In [GP02, Proposition 3.4] it is shown that if an element $x \in \mathbb{X}$ has finite support, then there is a unique least finite set of atoms that supports x.

Definition 5.2. When $x \in \mathbb{X}$ has finite support, call the least set of atoms that supports x the **support** of x, and write it as $\mathrm{supp}(x)$.

Write $a\#x$ when $a \notin \mathrm{supp}(x)$, and say that a **is fresh for** x.

Lemma 5.3. *Basic results on nominal sets are:*

1. $\mathrm{supp}(x) = \{a \in \mathbb{A} \mid \{b \in \mathbb{A} \mid (a\ b) \cdot x \neq x\}$ *is not finite*$\}$.
2. *If* $a\#x$ *for every* $a \in \mathrm{ds}(\pi, \pi')$ *then* $\pi \cdot x = \pi' \cdot x$.
3. *If* $a\#x$ *then* $\pi(a)\#\pi \cdot x$.

Proof. Elsewhere [GP02, Proposition 3.4] and by calculations. □

Remark 5.4. For two elements x and y of a set, $x = y$ denotes that x is the *same* element as y (this should not be confused with provable equality between terms $t = u$ from the previous sections, where this was not the case). Therefore properties like 'if $x = y$ then $\pi \cdot x = \pi \cdot y$' and 'if $x = y$ then $a\#x$ if and only if $a\#y$' trivially follow from the fact that a single element in a set has exactly the same properties as... itself.

Example 5.5.

1. The set $\mathbb{A}$ of all atoms with action $\pi \cdot a = \pi(a)$ is a nominal set; the support of $a \in \mathbb{A}$ is $\{a\}$. Note that for $x, y \in \mathbb{A}$, $x\#y$ when $x \neq y$.
2. The powerset $\mathcal{P}(\mathbb{A}) = \{U \mid U \subseteq \mathbb{A}\}$ of $\mathbb{A}$ with action $\pi \cdot U = \{\pi \cdot u \mid u \in U\}$, is *not* a nominal set; $\{a_1, a_3, a_5, \ldots\} \in \mathcal{P}(\mathbb{A})$ does not have finite support, since for no finite set of atoms it is the case that all permutations fixing that set map $\{a_1, a_3, a_5, \ldots\}$ to itself. Note that the support of $\mathbb{A} \in \mathcal{P}(\mathbb{A})$ is $\emptyset$, so $a\#\mathbb{A}$ for any a.
3. Call $U \subseteq \mathbb{A}$ **cofinite** when $\mathbb{A}\backslash U$ is finite. The set $\mathcal{P}_{fs}(\mathbb{A})$ of finite and cofinite subsets of $\mathbb{A}$, which can be defined as $\{U \mid U \subseteq \mathbb{A},\ U$ finite or cofinite$\}$, with the pointwise action inherited from $\mathcal{P}(\mathbb{A})$, is a nominal set (*fs* stands for finite support); the support of $\mathbb{A}\backslash\{a\} \in \mathcal{P}_{fs}(\mathbb{A})$ is a, so $b\#\mathbb{A}\backslash\{a\}$ but not $a\#\mathbb{A}\backslash\{a\}$.
4. The empty set $\emptyset$ with the trivial action is a nominal set.
5. If $\mathbb{X}$ and $\mathbb{Y}$ are nominal sets write $\mathbb{X} \times \mathbb{Y}$ for $\{(x, y) \mid x \in \mathbb{X},\ y \in \mathbb{Y}\}$ with action $\pi \cdot (x, y) = (\pi \cdot x, \pi \cdot y)$. This is also a nominal set; the support of $(x, y) \in \mathbb{X} \times \mathbb{Y}$ is the union of the supports of x and y.
6. If $\mathbb{X}$ is a nominal set write $\mathbb{X}^n$ for $\{(x_1, \ldots, x_n) \mid x_i \in \mathbb{X}, 1 \leq i \leq n\}$ with action $\pi \cdot (x_1, \ldots, x_n) = (\pi \cdot x_1, \ldots, \pi \cdot x_n)$. Again, this is a nominal set; the support of an element $(x_1, \ldots, x_n)$ is the union of the supports of the x_i.
7. The set of infinitary λ-terms [KKSdV97] with the pointwise action is *not* a nominal set: terms might mention an infinite number of different atoms, so they do not adhere to the finite support property. This problem can be overcome by moving to FMG (Fraenkel Mostowski Generalised). This

generalises the countable set of atoms to any large cardinality, and finite sets of atoms to any strictly smaller cardinality (well-orderable sets to be precise); see elsewhere [Gab07a] for further details.

We can think of $a\#x$ as an abstract notion of 'does not occur in x in any *distinguished* manner'. We say 'distinguished', because the example of $\mathcal{P}_{fs}(\mathbb{A})$ shows that $\#$ is not the same as $\notin$: for example $a \notin (\mathbb{A}\backslash\{a\})$ but *not* $a\#(\mathbb{A}\backslash\{a\})$.

Definition 5.6. We assume the permutation actions on the sets from Example 5.5 henceforth.

Definition 5.7. For any nominal sets $\mathbb{X}, \mathbb{Y}$, call a function $f \in \mathbb{X} \to \mathbb{Y}$ (on the underlying sets) **equivariant** when $\pi \cdot f(x) = f(\pi \cdot x)$ for any $x \in \mathbb{X}$.

Lemma 5.8. *For any nominal sets $\mathbb{X}, \mathbb{Y}$, equivariant function $f \in \mathbb{X} \to \mathbb{Y}$ and $x \in \mathbb{X}$,* $\mathrm{supp}(f(x)) \subseteq \mathrm{supp}(x)$.
As a corollary, $a\#x$ implies $a\#f(x)$.

Proof. By Definition 5.7 $\pi \cdot f(x) = f(\pi \cdot x)$, so if $\pi \cdot x = x$ then $\pi \cdot f(x) = f(x)$.
The corollary follows by Definition 5.2. □

Subsets of (the underlying set of) a nominal set will be important later when we build free term algebras.

Definition 5.9. $\mathcal{X} \subseteq \mathbb{X}$ inherits a pointwise action $\pi \cdot \mathcal{X} = \{\pi \cdot x \mid x \in \mathcal{X}\}$.
We will always use this action on $\mathcal{X} \subseteq \mathbb{X}$.

$a\#\mathcal{X}$ does *not* imply that $a\#x$ for every $x \in \mathcal{X}$. For example $\mathbb{A} \subseteq \mathbb{A}$ and it is a fact that $a\#\mathbb{A}$ — but $a \in \mathbb{A}$ and not $a\#a$. Furthermore $\mathcal{X} \subseteq \mathbb{X}$ does *not* imply that $\mathcal{X}$ is finitely supported. For example $\{a_1, a_2, a_3, \ldots\} \subseteq \mathbb{A}$ but $\{a_1, a_3, a_5, \ldots\}$ is not finitely supported. However, the finitely-supported subsets of $\mathbb{X}$ form a nominal set — they have a permutation action, and are finitely supported.

Lemma 5.10. *Suppose $\mathbb{X}$ is a nominal set and $\mathcal{X} \subseteq \mathbb{X}$ is finitely-supported. Then if $a_1\#\mathcal{X}, \ldots, a_n\#\mathcal{X}$ there exists some $x \in \mathcal{X}$ such that $a_1\#x, \ldots, a_n\#x$.*

Proof. Choose any $y \in \mathcal{X}$. Let $b_1, \ldots, b_n$ be fresh (so $b_i\#\mathcal{X}$ and $b_i\#y$ for $1 \leq i \leq n$). Then by part 2 of Lemma 5.3 $(b_1\ a_1) \cdots (b_n\ a_n) \cdot \mathcal{X} = \mathcal{X}$.
Write $x = (b_1\ a_1) \cdots (b_n\ a_n) \cdot y$. Then $x \in \mathcal{X}$ by Definition 5.9 and we conclude $a_i\#x$ for $1 \leq i \leq n$ by part 3 of Lemma 5.3 and the assumption $b_i\#y$. □

6 Interpretations, Models and Validity

We now give a semantics in nominal sets to nominal algebra theories.

Definition 6.1. An **interpretation of a signature** Σ, which we write as $[\![_]\!]$, is a nominal set $\mathbb{T}$ with *equivariant* functions

- $[\![_]\!] \in \mathbb{A} \to \mathbb{T}$ to interpret atoms;
- $[_]_ \in \mathbb{A} \times \mathbb{T} \to \mathbb{T}$ such that $a\#[a]x$ always, to interpret abstraction;
- $[\![\mathsf{f}]\!] \in \mathbb{T}^n \to \mathbb{T}$ for each term-former $\mathsf{f} : n$ in Σ, to interpret term-formers.

We extend the notion of interpretation to terms, where we map unknowns to elements of the nominal set $\mathbb{T}$:

Definition 6.2. A **valuation** ς maps unknowns X to elements $\varsigma(X) \in \mathbb{T}$. We write $[\![t]\!]_\varsigma$ for the **interpretation of a term** t under a valuation ς, inductively defined by:

$$[\![a]\!]_\varsigma = [\![a]\!] \qquad [\![\pi \cdot X]\!]_\varsigma = \pi \cdot \varsigma(X) \qquad [\![[a]t]\!]_\varsigma = [a][\![t]\!]_\varsigma$$
$$[\![\mathsf{f}(t_1, \ldots, t_n)]\!]_\varsigma = [\![\mathsf{f}]\!]([\![t_1]\!]_\varsigma, \ldots, [\![t_n]\!]_\varsigma)$$

Interpretations are equivariant.

Lemma 6.3. *For any* π, $\pi \cdot [\![t]\!]_\varsigma = [\![\pi \cdot t]\!]_\varsigma$.

Proof. By induction on the structure of t, using Lemma 5.8 for the cases of a, $[a]t$ and $\mathsf{f}(t_1, \ldots, t_n)$. □

Using the interpretations of signatures and terms, we define the notion of *validity* on judgement forms as follows:

Definition 6.4. For any interpretation $[\![_]\!]$, say that:

$$[\![\Delta]\!]_\varsigma \text{ (**is valid**) when } a\#\varsigma(X) \text{ for each } a\#X \in \Delta$$
$$[\![\Delta \vdash a\#t]\!]_\varsigma \text{ when } [\![\Delta]\!]_\varsigma \text{ implies } a\#[\![t]\!]_\varsigma$$
$$[\![\Delta \vdash t = u]\!]_\varsigma \text{ when } [\![\Delta]\!]_\varsigma \text{ implies } [\![t]\!]_\varsigma = [\![u]\!]_\varsigma$$
$$[\![\Delta \vdash a\#t]\!] \text{ when } [\![\Delta \vdash a\#t]\!]_\varsigma \text{ for all valuations } \varsigma$$
$$[\![\Delta \vdash t = u]\!] \text{ when } [\![\Delta \vdash t = u]\!]_\varsigma \text{ for all valuations } \varsigma$$

Then a model of a theory is an interpretation that validates its axioms:

Definition 6.5. A **model** of a theory T is an interpretation $[\![_]\!]$ of its signature such that $[\![\nabla \vdash t = u]\!]$ for all axioms $\nabla \vdash t = u$ of T.

So a model of a nominal algebra theory is just like a model of any other algebraic theory, but we must interpret permutations by permutations, and atoms, abstractions and term-formers by *equivariant* functions on the underlying sets.

Definition 6.6. For any theory T, define **validity with respect to** T for judgement forms as follows:

- Write $\Delta \models_\mathsf{T} a\#t$ when $[\![\Delta \vdash a\#t]\!]$ for all models $[\![_]\!]$ of T.
- Write $\Delta \models_\mathsf{T} t = u$ when $[\![\Delta \vdash t = u]\!]$ for all models $[\![_]\!]$ of T.

Note the T subscript in $\models_{\mathsf{T}}$, which indicates that freshness validity is not a purely syntactic affair as in freshness derivability, but also depends on the axioms of theory T. More on this in Subsection 7.3.

Derivability of freshness and equality is sound for the semantics:

Theorem 6.7 (Soundness)**.** *For any* T*,* Δ*,* a*,* t, u*:*

1. *If* $\Delta \vdash a\#t$ *then* $\Delta \models_{\mathsf{T}} a\#t$*.*
2. *If* $\Delta \vdash_{\mathsf{T}} t = u$ *then* $\Delta \models_{\mathsf{T}} t = u$*.*

Proof. Let $[\![_]\!]$ be a model of T. We must show that if $a\#t$ (or $t = u$) is derived from Δ then $[\![\Delta]\!]_\varsigma$ implies $a\#[\![t]\!]_\varsigma$ (or $[\![t]\!]_\varsigma = [\![u]\!]_\varsigma$) for any valuation ς. We work by induction on derivations built using the rules in Figures 1 and 2.

- (#**ab**). We must show $a\#[\![b]\!]$. By Lemma 5.8 this follows from $a\#b$, which is a standard property of freshness (see part 1 of Example 5.5).
- (#**X**). By the inductive hypothesis we know $\pi^{-1}(a)\#\varsigma(X)$. By part 3 of Lemma 5.3 we conclude $a\#\pi \cdot \varsigma(X)$.
- (#[]**a**). $a\#[a][\![t]\!]_\varsigma$ holds by construction.
- (#[]**b**). $a\#[\![t]\!]_\varsigma$ implies $a\#[b][\![t]\!]_\varsigma$, by Lemma 5.8.
- (#**f**). If $a\#[\![t_i]\!]_\varsigma$ for $1 \leq i \leq n$ then $a\#[\![\mathsf{f}]\!]([\![t_1]\!]_\varsigma, \ldots, [\![t_n]\!]_\varsigma)$ follows using Lemma 5.8.
- (**refl**), (**symm**), (**tran**), (**cong**[]), (**congf**). By properties of equality.
- (**perm**). We know that $a\#[\![t]\!]_\varsigma$ and $b\#[\![t]\!]_\varsigma$ imply $(a\ b) \cdot [\![t]\!]_\varsigma = [\![t]\!]_\varsigma$ by part 2 of Lemma 5.3. We conclude $[\![(a\ b) \cdot t]\!]_\varsigma = [\![t]\!]_\varsigma$ by Lemma 6.3.
- (**ax**$_{\nabla \vdash \mathbf{t=u}}$). Suppose $[\![\nabla^\pi \sigma]\!]_\varsigma$ for any ς. Then $\pi(a)\#\sigma(X)\varsigma$ holds for all $a\#X \in \nabla$. By part 3 of Lemma 5.3 also $a\#\pi^{-1} \cdot \sigma(X)\varsigma$ for all $a\#X \in \nabla$. Let ς' be defined as $\varsigma'(X) = \pi^{-1} \cdot \sigma(X)\varsigma$ for any X. Then $a\#\varsigma'(X)$ for all $a\#X \in \nabla$, so $[\![\nabla]\!]_{\varsigma'}$ holds. Since $\nabla \vdash t = u$ is an axiom of T, we know $[\![t]\!]_{\varsigma'} = [\![u]\!]_{\varsigma'}$. Then trivially also $\pi \cdot [\![t]\!]_{\varsigma'} = \pi \cdot [\![u]\!]_{\varsigma'}$, and by Lemma 6.3 we obtain $[\![\pi \cdot t]\!]_{\varsigma'} = [\![\pi \cdot u]\!]_{\varsigma'}$. Now by a straightforward induction on syntax we can verify that $[\![\pi \cdot t]\!]_{\varsigma'} = [\![t^\pi \sigma]\!]_\varsigma$ and $[\![\pi \cdot u]\!]_{\varsigma'} = [\![u^\pi \sigma]\!]_\varsigma$, and we conclude $[\![t^\pi \sigma]\!]_\varsigma = [\![u^\pi \sigma]\!]_\varsigma$.
- (**fr**). So suppose $\Delta \vdash_{\mathsf{T}} t = u$ is derived from $\Delta, a\#X_1, \ldots, a\#X_n \vdash_{\mathsf{T}} t = u$, where $a \notin t, u, \Delta$. By equivariance (Theorem A.4) then also

  $$\Delta, a'\#X_1, \ldots, a'\#X_n \vdash_{\mathsf{T}} t = u$$

  for all other a' not occurring in Δ, t or u. We also retain the inductive hypothesis for $\Delta, a'\#X_1, \ldots, a'\#X_n \vdash_{\mathsf{T}} t = u$ by ZFA equivariance.
  We must show that $[\![\Delta \vdash_{\mathsf{T}} t = u]\!]_\varsigma$ for any ς. Now pick an $a' \notin \Delta, t, u$ such that $a'\#\varsigma(X_i)$ for $1 \leq i \leq n$. Then by the inductive hypothesis we obtain $[\![\Delta, a'\#X_1, \ldots, a'\#X_n \vdash_{\mathsf{T}} t = u]\!]_\varsigma$. But this is equivalent to $[\![\Delta \vdash_{\mathsf{T}} t = u]\!]_\varsigma$, since $[\![a'\#X_1, \ldots, a'\#X_n]\!]_\varsigma$. The result follows.

□

7 Completeness

In this section we show that derivability of equality is complete with respect to the semantics. We also show that completeness does not hold for derivability of freshness, and that this is precisely what we want. For this we use the notion of a free term model, which we introduce first.

7.1 Free Term Models

Definition 7.1. In this subsection fix a signature Σ, a theory $\mathsf{T} = (\Sigma, Ax)$, and fix a set of term-formers D disjoint from Σ.

The usual technique to obtain models for a theory T is to add constant symbols to the language (to ensure a supply of 'arbitrary elements') and quotient by provable equality. But in nominal algebra constants have empty support; if d has arity 0 then $\vdash a\#\mathsf{d}$ is derivable for any a. Adding constants only ensures a supply of elements with empty support.

To reflect in syntax that an element of a nominal set can have support, we use n-ary term-formers d applied to n distinct atoms. This idea goes back to [Gab06]. We now give the construction in detail.

Definition 7.2. Let **free terms** be inductively generated by the following grammar:

$$g \quad ::= \quad a \;\mid\; [a]g \;\mid\; \mathsf{f}(g_1, \ldots, g_n) \;\mid\; \mathsf{d}(a_1, \ldots, a_n)$$

Here $\mathsf{f} : n$ ranges over elements of Σ, and $\mathsf{d} : n$ ranges over elements of D.

Recall the notation $\pi \cdot t$ for the object-level permutation action on a term t from Definition 3.1.

Lemma 7.3. *The set of free terms with action $\pi \cdot g$ is a nominal set; the support of g is $\{a \in \mathbb{A} \mid a \notin g\}$.*

As a corollary, $a \notin g$ if and only if $a\#g$.

Proof. $\mathrm{supp}(g) = \{a \in \mathbb{A} \mid a \notin g\}$ follows by an induction on the structure of g, using part 1 of Lemma 5.3. The corollary follows by Definition 5.2. □

We need a technical lemma:

Lemma 7.4. *For any free term g, if $a \notin g$ then $\vdash a\#g$.*

Proof. By an induction on syntax using the rules in Figure 1. □

Definition 7.5. Write (**cong**d) for an instance of the (**cong**f) rule when $\mathsf{f} \in \mathsf{D}$. Write $[g]_{\mathsf{T}}$ for the set of free terms g' such that a derivation of $\vdash_{\mathsf{T}} g = g'$ exists that does not mention (**cong**d) for any $\mathsf{d} \in \mathsf{D}$.

Let the **set of free terms up to** T be the set $\{[g]_{\mathsf{T}} \mid g \text{ a free term}\}$.

Lemma 7.6. *The set of free terms up to* T *with action* $\pi \cdot [g]_{\mathsf{T}} = [\pi \cdot g]_{\mathsf{T}}$ *is a nominal set; the set* $[g]_{\mathsf{T}}$ *is supported by* $\{a \in \mathbb{A} \mid \nvdash a\#g\}$.

As a corollary, if $\vdash a\#g$ *then* $a\#[g]_{\mathsf{T}}$.

Proof. Let A be the set $\{a \mid \nvdash a\#g\}$; this is finite by Lemma 7.4. It suffices to show that A supports $[g]_{\mathsf{T}}$.

So let π be a permutation such that $\pi(a) = a$ for all $a \in A$. We must show $\pi \cdot [g]_{\mathsf{T}} = [g]_{\mathsf{T}}$. By assumption $\pi \cdot [g]_{\mathsf{T}} = [\pi \cdot g]_{\mathsf{T}}$, so it suffices to show $[\pi \cdot g]_{\mathsf{T}} = [g]_{\mathsf{T}}$. By definition of $[_]_{\mathsf{T}}$, this is when $\vdash_{\mathsf{T}} \pi \cdot g = g$. By Lemma 4.13, this follows from $\vdash \mathrm{ds}(\pi, \iota)\#g$. But this follows from the assumption on π since $\mathrm{ds}(\pi, \iota)$ and A are disjoint.

The corollary follows by Definition 5.2. □

The following technical lemma will be useful later:

Lemma 7.7. $a_1\#[g]_{\mathsf{T}}, \ldots, a_n\#[g]_{\mathsf{T}}$ *if and only if there exists some* $g' \in [g]_{\mathsf{T}}$ *such that* $\vdash a_1\#g', \ldots, \vdash a_n\#g'$ *are all derivable.*

Proof. For the left-right implication we use Lemma 5.10 to pick some $g' \in [g]_{\mathsf{T}}$ such that $a_1\#g', \ldots, a_n\#g'$. We conclude $\vdash a_1\#g', \ldots, \vdash a_n\#g'$ by Lemmas 7.3 and 7.4.

For the right-to-left implication, we observe that if $g' \in [g]_{\mathsf{T}}$ then $[g]_{\mathsf{T}} = [g']_{\mathsf{T}}$. The result follows by Lemma 7.6. □

The following example shows why Lemma 7.7 is non-trivial.

Example 7.8. Consider a theory ATOM with one axiom $\vdash a = b$. It is easy to verify that $a\#[a]_{\mathsf{ATOM}}$ (since $[a]_{\mathsf{ATOM}} = \mathbb{A}$) but $a\#a$ is not derivable. Of course $a = b$ and $a\#b$ *are* derivable. Similarly in LAM it is a fact that $a\#[(\lambda[a]b)a]_{\mathsf{LAM}}$ but $a\#(\lambda[a]b)a$ is not derivable; of course $(\lambda[a]b)a = b$ and $a\#b$ are derivable.

Definition 7.9. We construct the **free term model** $[\![_]\!]^{\mathcal{T}}$ of T over D as follows:

- Take as underlying nominal set the set of free terms up to T with action $\pi \cdot [g]_{\mathsf{T}} = [\pi \cdot g]_{\mathsf{T}}$.
- Take as interpretations of the equivariant functions:
  - $[\![a]\!]^{\mathcal{T}} = [a]_{\mathsf{T}}$.
  - $[a]x = [[a]g]_{\mathsf{T}}$ for some $g \in x$.
  - $[\![\mathsf{f}]\!]^{\mathcal{T}}(x_1, \ldots, x_n) = [\mathsf{f}(g_1, \ldots, g_n)]_{\mathsf{T}}$ for some $g_1 \in x_1, \ldots, g_n \in x_n$ (for each term-former $\mathsf{f} : n$ in Σ).

It is usual to build free term models by quotienting by provable equality; we exclude (**congd**) to avoid the following degenerate case: if we allow (**congd**) and T contains an axiom such as $\vdash a = b$, then $\mathrm{supp}[\mathsf{d}(a_1, \ldots, a_n)]_{\mathsf{T}} = \emptyset$. This is not the behaviour we want. Similarly our syntax of free terms does not allow terms of the form $\mathsf{d}(g_1, \ldots, g_n)$ in general. The only purpose of $[\mathsf{d}(a_1, \ldots, a_n)]_{\mathsf{T}}$ is to 'be an unknown element with support $a_1, \ldots, a_n$'.

Lemma 7.10. $[\![_]\!]^{\mathcal{T}}$ *is an interpretation of* Σ.

Proof. The underlying set of $[\![_]\!]^{\mathcal{T}}$ is a nominal set by Lemma 7.6. For the interpretation functions we must show that they are well-defined (for $[_]_$ and $[\![\mathsf{f}]\!]^{\mathcal{T}}$ the choices of $g \in x$ and $g_1 \in x_1, \ldots, g_n \in x_n$ do not matter) and equivariant. This is easy using the definitions of $[_]_{\mathsf{T}}$ and the permutation action. Finally, for the $[_]_$ function, we must show that $a\#[a]x$ holds, i.e. $a\#[[a]g]_{\mathsf{T}}$, which we will do now.

Choose b fresh (so $b \notin g$ and $b\#[[a]g]_{\mathsf{T}}$). Since $b\#[[a]g]_{\mathsf{T}}$, also $a\#(b\ a) \cdot [[a]g]_{\mathsf{T}}$ by part 3 of Lemma 5.3. By definition of the permutation action, then also $a\#[[b](b\ a) \cdot g]_{\mathsf{T}}$. Now since $\vdash a\#[b](b\ a) \cdot g$ and $\vdash b\#[b](b\ a) \cdot g$ by Lemma 7.4 and the derivation rules of Figure 1, we know $\vdash_{\mathsf{T}} [b](b\ a) \cdot g = [a]g$ by (**perm**). Then $[[b](b\ a) \cdot g]_{\mathsf{T}} = [[a]g]_{\mathsf{T}}$, and we obtain $a\#[[a]g]_{\mathsf{T}}$ as required. □

Theorem 7.11. $[\![_]\!]^{\mathcal{T}}$ *is a model of* T.

Proof. Suppose $\nabla \vdash t = u$ is an axiom of T. Suppose that ς is a valuation to the underlying set of $[\![_]\!]^{\mathcal{T}}$ and that $a\#\varsigma(X)$ for every $a\#X \in \nabla$. We must show that $[\![t]\!]_\varsigma = [\![u]\!]_\varsigma$.

Let $\mathcal{X}$ be the set of all unknowns mentioned in ∇, t, or u. By Lemma 7.7, for every $X \in \mathcal{X}$ there is an element $g_X \in \varsigma(X)$ such that $\vdash a\#g_X$ for every $a\#X \in \nabla$. Let σ be the substitution such that $\sigma(X) = g_X$ when $X \in \mathcal{X}$ and $\sigma(X) = X$ when $X \notin \mathcal{X}$. Then $\vdash \nabla\sigma$, and $\vdash_{\mathsf{T}} t\sigma = u\sigma$ by $(\mathbf{ax}_{\nabla \vdash \mathbf{t}=\mathbf{u}})$. Since this derivation does not mention (**congd**), we know $[t\sigma]_{\mathsf{T}} = [u\sigma]_{\mathsf{T}}$ by Definition 7.5. By an induction on syntax we verify that $[t\sigma]_{\mathsf{T}} = [\![t]\!]_\varsigma$ and $[u\sigma]_{\mathsf{T}} = [\![u]\!]_\varsigma$, and the result follows. □

7.2 Completeness for Equality Derivations

Definition 7.12. For this subsection, fix a signature Σ, a theory $\mathsf{T} = (\Sigma, Ax)$, and terms t, u and a freshness context Δ in signature Σ.

We will show that derivability of $\Delta \vdash_{\mathsf{T}} t = u$ is complete. That is, we will prove:

Theorem 7.13 (Completeness)**.** *If* $\Delta \models_{\mathsf{T}} t = u$ *then* $\Delta \vdash_{\mathsf{T}} t = u$.

The proof takes up the rest of this subsection.

We shall consider a specific free term model and a specific valuation to this model that preserves sufficient information to allow us to reconstruct a derivation of $\Delta \vdash_{\mathsf{T}} t = u$.

Definition 7.14. Let $\mathcal{X}$ be the unknowns mentioned in Δ, t, u, and let $\mathcal{A}$ be the atoms mentioned in Δ, t, u. For each $X \in \mathcal{X}$:

- let $a_{X1}, \ldots, a_{Xk_X}$ be the atoms in $\mathcal{A}$ (in some arbitary but fixed order) such that $a_{Xi}\#X \notin \Delta$;
- let $\mathsf{d}_X : k_X$ be a term-former.

For each unknown $X \notin \mathcal{X}$, let $\mathsf{d}_X : 0$ be a term-former.

Let D be the set of all d_X's (so $\mathsf{d}_X \in \mathsf{D}$ for each X).

Definition 7.15. Let σ be the following substitution:

$$\begin{array}{ll} \sigma(X) = \mathsf{d}_X(a_{X1}, \ldots, a_{X k_X}) & (X \in \mathcal{X}) \\ \sigma(X) = \mathsf{d}_X() & (X \notin \mathcal{X}) \end{array}$$

We consider the free term model $[\![_]\!]^{\mathcal{T}}$ of T over D, and valuation ς such that $\varsigma(X) = [\sigma(X)]_{\mathsf{T}}$.

Lemma 7.16. *$[t\sigma]_{\mathsf{T}} = [\![t]\!]_\varsigma$ and $[u\sigma]_{\mathsf{T}} = [\![u]\!]_\varsigma$.*

Proof. By an easy induction on syntax. □

Lemma 7.17. *$[\![\Delta]\!]_\varsigma$ holds.*

Proof. Suppose $a\#X \in \Delta$. We must show that $a\#\varsigma(X)$. By construction $X \in \mathcal{X}$ so $\varsigma(X) = [\mathsf{d}_X(a_{X1}, \ldots, a_{X k_X})]_{\mathsf{T}}$. But also $a \notin \{a_{X1}, \ldots, a_{X k_X}\}$ by construction so $a \notin \mathsf{d}_X(a_{X1}, \ldots, a_{X k_X})$. The result follows by Lemmas 7.4 and 7.6. □

Definition 7.18. Let Π be a derivation of $\vdash_{\mathsf{T}} t\sigma = u\sigma$ without using (**congd**). By Corollary 4.9 we assume that Π does not contain unknowns or instances of (#**X**) and (**fr**).

Let $\mathcal{A}^+$ be $\mathcal{A}$ extended with:

- atoms mentioned anywhere in Π (that were not already in $\mathcal{A}$);
- a set $\mathcal{B} = \{b_{Xi} \mid X, i \text{ such that } a_{Xi} \in \mathcal{A}\}$ of fresh atoms in bijection with $\mathcal{A}$;
- one fresh atom c (so c does not occur in $\mathcal{A}$, Π, or $\mathcal{B}$).

Let Δ^+ be Δ extended with freshness assumptions $a'\#X$ for every $X \in \mathcal{X}$ and every $a' \in \mathcal{A}^+ \setminus \mathcal{A}$.

Definition 7.19. For the rest of this subsection let g and h range over free terms in $\Sigma \cup \mathsf{D}$ that mention only atoms from $\mathcal{A}^+ \setminus (\mathcal{B} \cup \{c\})$. Define an **inverse mapping** -^{-1} from such free terms to terms in Σ inductively as follows, where $X \in \mathcal{X}$ and $Y \notin \mathcal{X}$, and where we omit $\circ$ between compositions of swappings, for brevity:

$$\begin{array}{c} a^{-1} \equiv a \qquad ([a]g)^{-1} \equiv [a]g^{-1} \qquad \mathsf{f}(g_1, \ldots, g_n)^{-1} \equiv \mathsf{f}(g_1^{-1}, \ldots, g_n^{-1}) \\ \mathsf{d}_X(a'_{X1}, \ldots, a'_{X k_X})^{-1} \equiv (a'_{X1}\ b_{X1}) \cdots (a'_{X k_X}\ b_{X k_X})(b_{X1}\ a_{X1}) \cdots (b_{X k_X}\ a_{X k_X}) \cdot X \\ \mathsf{d}_Y()^{-1} \equiv c \end{array}$$

We extend the notation -^{-1} to freshnesses and freshness contexts by acting on the terms they mention.

The inverse mapping is equivariant (for the terms we care about):

Lemma 7.20. *$\Delta^+ \vdash_{\mathsf{CORE}} (\pi \cdot g)^{-1} = \pi \cdot g^{-1}$ when the atoms that π mentions are all from $\mathcal{A}^+ \setminus (\mathcal{B} \cup \{c\})$.*

Proof. By induction on the structure of g. The only non-trivial case is when $g \equiv \mathsf{d}_X(a'_{X1}, \ldots, a'_{Xk_X})$ with $X \in \mathcal{X}$. Then we must show

$$\Delta^+ \vdash_{\mathsf{CORE}} \pi' \cdot X = \pi'' \cdot X,$$

where we used the following abbreviations:

$$\begin{aligned}\pi' &= (\pi(a'_{X1})\ b_{X1}) \cdots (\pi(a'_{Xk_X})\ b_{Xk_X})(b_{X1}\ a_{X1}) \cdots (b_{Xk_X}\ a_{Xk_X}) \\ \pi'' &= \pi \circ (a'_{X1}\ b_{X1}) \cdots (a'_{Xk_X}\ b_{Xk_X})(b_{X1}\ a_{X1}) \cdots (b_{Xk_X}\ a_{Xk_X})\end{aligned}$$

By the syntactic criteria for CORE derivability (Corollary 4.16) it suffices to show $\Delta^+ \vdash \mathrm{ds}(\pi', \pi'')\#X$. The result follows by a case analysis on the atoms in the difference set, using the fact that π does not mention any of the b_{X_i}. □

Lemma 7.21. *$\Delta^+ \vdash_{\mathsf{CORE}} t\sigma^{-1} = t$ and $\Delta^+ \vdash_{\mathsf{CORE}} u\sigma^{-1} = u$.*

Proof. We show $\Delta^+ \vdash_{\mathsf{CORE}} v\sigma^{-1} = v$ for each subterm v of t and u. We do this by induction on the structure of v. The proof of the case of $v \equiv \pi \cdot X$ is analogous to the $\mathsf{d}_X(a'_{X1}, \ldots, a'_{Xk_X})$ case in the proof of Lemma 7.20. □

Lemma 7.22. *If $\vdash_{\mathsf{T}} t\sigma = u\sigma$ without using* (**cong**d) *then $\Delta \vdash_{\mathsf{T}} t = u$.*

Proof. Suppose we could transform the derivation Π of $\vdash_{\mathsf{T}} t\sigma = u\sigma$ into a derivation of $\Delta^+ \vdash_{\mathsf{T}} t\sigma^{-1} = u\sigma^{-1}$. Given that, the result follows because by Lemma 7.21 we deduce $\Delta^+ \vdash_{\mathsf{T}} t = u$ and we obtain $\Delta \vdash_{\mathsf{T}} t = u$ as required by a number of applications of (**fr**).

Our transformation of $\vdash_{\mathsf{T}} t\sigma = u\sigma$ into $\Delta^+ \vdash_{\mathsf{T}} t\sigma^{-1} = u\sigma^{-1}$ is inductive on Π. Suppose Π concludes with an instance of ...

- (#**ab**), (#[**a**]), (#[**b**]), (**refl**), (**symm**), (**tran**) or (**cong**[]). Then the result trivially follows by an instance of the same rule, possibly using the inductive hypothesis.
- (#**X**) or (**fr**). This is impossible by assumption (see Definition 7.18).
- (#**f**). There are three cases to consider:
  - The case of $\vdash a\#\mathsf{f}(g_1, \ldots, g_n)$ for $\mathsf{f} \in \Sigma$.
    Then by assumption $\vdash a\#g_i$ for $1 \leq i \leq n$, and $\Delta^+ \vdash a\#g_i^{-1}$ by the inductive hypothesis. We conclude $\Delta^+ \vdash a\#\mathsf{f}(g_1^{-1}, \ldots, g_n^{-1})$ using (#**f**).
  - The case of $\vdash a\#\mathsf{d}_X(a'_{X1}, \ldots, a'_{Xk_X})$ for $\mathsf{d}_X \in \mathsf{D}$ and $X \in \mathcal{X}$.
    Then by assumption $\vdash a\#a'_{X_i}$ for $1 \leq i \leq k_X$, and we obtain $a \neq a'_{X_i}$ by (#**ab**). We must show $\Delta^+ \vdash a\#\pi \cdot X$, where

    $$\pi = (a'_{X1}\ b_{X1}) \cdots (a'_{Xk_X}\ b_{Xk_X})(b_{X1}\ a_{X1}) \cdots (b_{Xk_X}\ a_{Xk_X}).$$

    By (#**X**), this follows from $\Delta^+ \vdash \pi^{-1}(a)\#X$. Since $a \neq a'_{X_i}$ and also $a \neq b_{X_i}$ for all i, we have $\pi^{-1}(a) = (b_{X1}\ a_{X1}) \cdots (b_{Xk_X}\ a_{Xk_X})(a)$. We proceed by a case distinction on a:

  - ∗ If $a = a_{X_i}$ for some i, then $(b_{X_1}\ a_{X_1}) \cdots (b_{X_{k_X}}\ a_{X_{k_X}})(a) = b_{X_i}$, and the result follows since $b_{X_i} \# X \in \Delta^+$ by construction.
  - ∗ If $a \neq a_{X_i}$ for all i, then $(b_{X_1}\ a_{X_1}) \cdots (b_{X_{k_X}}\ a_{X_{k_X}})(a) = a$ since also $a \neq b_{X_j}$ for any j. Then by construction $a \# X \in \Delta$ since the a_{X_i} are the only atoms in $\mathcal{A}$ for which $a_{X_i} \# X \notin \Delta$. The result follows.
- The case of $\vdash a \# \mathsf{d}_X()$ for $\mathsf{d}_X \in \mathsf{D}$ and $X \notin \mathcal{X}$.
  It is immediate by (**#ab**) that $\vdash a \# c$.

- (**congf**). We consider two cases:
  - The case of $\mathsf{f} \in \Sigma$ follows using the inductive hypothesis.
  - The case of $\mathsf{d} \in \mathsf{D}$ is impossible, since we assumed that Π does not mention (**congd**).
- (**perm**). By the inductive hypothesis we have $\Delta^+ \vdash a \# g^{-1}$ and $\Delta^+ \vdash b \# g^{-1}$. Then $\Delta^+ \vdash_{\mathsf{T}} (a\ b) \cdot g^{-1} = g^{-1}$ by (**perm**). Using Lemma 7.20, we conclude $\Delta^+ \vdash_{\mathsf{T}} ((a\ b) \cdot g)^{-1} = g^{-1}$.
- ($\mathbf{ax}_{\nabla \vdash \mathbf{v}=\mathbf{w}}$). Then $\vdash \nabla^\pi \tau$ and $\vdash_{\mathsf{T}} v^\pi \tau = w^\pi \tau$ for some permutation π and substitution τ such that $\nabla\tau$, $v\tau$ and $w\tau$ do not mention any unknowns. We must show $\Delta^+ \vdash_{\mathsf{T}} (v^\pi \tau)^{-1} = (w^\pi \tau)^{-1}$.
  Now let τ' be the substitution such that $\tau'(X) = \tau(X)^{-1}$ when $\tau(X) \neq X$ and $\tau'(X) = X$ when $\tau(X) = X$. Then $(v^\pi\tau)^{-1} \equiv v^\pi \tau'$, $(w^\pi\tau)^{-1} \equiv w^\pi \tau'$ and $(\nabla^\pi \tau)^{-1} \equiv \nabla^\pi \tau'$, so it suffices to show $\Delta^+ \vdash_{\mathsf{T}} v^\pi \tau' = w^\pi \tau'$. By ($\mathbf{ax}_{\nabla \vdash \mathbf{v}=\mathbf{w}}$), this follows from $\Delta^+ \vdash \nabla^\pi \tau'$, i.e. $\Delta^+ \vdash (\nabla^\pi \tau)^{-1}$. By the inductive hypothesis, this follows from the assumption $\vdash \nabla^\pi \tau$.

□

We are now ready for the main result of this subsection:

Proof of Theorem 7.13. Suppose $\Delta \models_{\mathsf{T}} t = u$, so $[\![\Delta \vdash t = u]\!]_\varsigma$ for the free term model $[\![_]\!]^{\mathcal{T}}$ and valuation ς constructed above. Now $[\![\Delta]\!]_\varsigma$ by Lemma 7.17 so $[\![t]\!]_\varsigma = [\![u]\!]_\varsigma$. By Lemma 7.16 $[t\sigma]_{\mathsf{T}} = [\![t]\!]_\varsigma$ and $[u\sigma]_{\mathsf{T}} = [\![u]\!]_\varsigma$. Therefore by construction $\vdash_{\mathsf{T}} t\sigma = u\sigma$ without using (**congd**). It follows by Lemma 7.22 that $\Delta \vdash_{\mathsf{T}} t = u$. □

7.3 The Status of Freshness Derivations

Definition 6.6 states that validity of freshness depends on the axioms of a theory. Derivability does not depend on axioms, it only inspects syntax. For this reason, we do not have completeness for freshnesses. That is, $\Delta \models_{\mathsf{T}} b \# t$ does *not* imply $\Delta \vdash b \# t$ necessarily.

For examples of incompleteness of freshness, we recall Example 7.8. We showed that in theory ATOM with one axiom $\vdash a = b$, $a \# [\![a]\!]^{\mathcal{T}}_\varsigma$ is valid for the term model $[\![_]\!]^{\mathcal{T}}$ (and any valuation ς). It is not hard to verify that this property holds for *all* models $[\![_]\!]$ of ATOM. So we have $\models_{\mathsf{ATOM}} a \# a$, but also $\nvdash a \# a$. Similarly, we can

show that $[\![(\lambda[a]b)a]\!]_\varsigma = [\![b]\!]_\varsigma$ for any model $[\![_]\!]$ of theory LAM, so $\models_{\mathsf{LAM}} a\#(\lambda[a]b)a$, but also $\not\vdash a\#(\lambda[a]b)a$.

To understand why this is desirable we must draw a distinction between the intension and the extension of a term.

'$\vdash a\#(\lambda[a]b)a$' has the status of '$x \notin fv((\lambda x.y)x)$'; both are false. Yet $(\lambda x.y)x$ is β-convertible to y and $x \notin fv(y)$ holds. A freshness judgement $\Delta \vdash a\#t$ is an intensional judgement on concrete syntax.[4] All the capture-avoidance side-conditions we know of are in accordance with the slogan 'ϵ away from informal practice', this is what $\Delta \vdash a\#t$ models.

Nominal sets is unusual amongst semantics in that it has an extensional *semantic* notion of freshness $a\#x$. In fact, semantic freshness is hiding in nominal algebra in the theory of *equality*.

Theorem 7.23. *Suppose $\{X_1, \ldots, X_n\}$ and $\{a_1, \ldots, a_m\}$ are the unknowns and atoms mentioned in Δ and t, and suppose that $b \notin \{a_1, \ldots, a_m\}$. Then:*

$$\Delta \models_{\mathsf{T}} a\#t \quad \textit{if and only if} \quad \Delta, b\#X_1, \ldots, b\#X_n \vdash_{\mathsf{T}} (b\ a) \cdot t = t.$$

Proof. Choose a model $[\![_]\!]$ and any valuation ς such that $b\#\varsigma(X_1), \ldots, b\#\varsigma(X_n)$; it follows by an induction on syntax that $b\#[\![t]\!]_\varsigma$. It is a fact that $a\#[\![t]\!]_\varsigma$ if and only if $(b\ a) \cdot [\![t]\!]_\varsigma = [\![t]\!]_\varsigma$ (see [GP02] for details). By Lemma 6.3, the last part is equivalent to $[\![(b\ a) \cdot t]\!]_\varsigma = [\![t]\!]_\varsigma$.

Now suppose $\Delta \models_{\mathsf{T}} a\#t$. By definition $a\#[\![t]\!]_\varsigma$ for any $[\![_]\!]$ and ς such that $[\![\Delta]\!]_\varsigma$. By the arguments above $[\![(b\ a) \cdot t]\!]_\varsigma = [\![t]\!]_\varsigma$ if $b\#\varsigma(X_1), \ldots, b\#\varsigma(X_n)$. By completeness Theorem 7.13 it follows that $\Delta, b\#X_1, \ldots, b\#X_n \vdash_{\mathsf{T}} (b\ a) \cdot t = t$. The reverse implication is similar. □

Theorem 7.23 tells us for instance that $\models_{\mathsf{ATOM}} a\#a$ if and only if $\vdash_{\mathsf{ATOM}} b = a$ (which follows by the axiom $\vdash a = b$), and that $\models_{\mathsf{LAM}} a\#(\lambda[a]b)a$ if and only if $\vdash_{\mathsf{LAM}} (\lambda[a]b)a = (\lambda[c]b)c$ (which follows since both sides are derivably equal to b).

So semantic freshness $\Delta \models_{\mathsf{T}} a\#t$ can be expressed as an equality axiom. Any undecidability or algorithmic complexity is isolated in the equality judgement form.

8 Conclusions

Nominal terms embrace the difference between the object-level and the meta-level. There are two classes of variables, atoms a and unknowns X. Substitution for X can capture abstractions by a. For instance the syntactic equality

$$(\lambda[a]X)[a/X] \equiv \lambda[a]a$$

formally reflects informal practice where instantiation in $\lambda x.t$ of t to x yields $\lambda x.x$. Freshness side-conditions of the form $a\#X$ manage the interaction between the two levels and 'recover' capture-avoidance where this is required.

[4] The reader familiar with a theorem-prover such as Isabelle [Pau89] might like to imagine that $\#$ maps to *Prop* and $=$ maps to o.

Nominal algebra stakes a claim to nominal terms as a *logical* system, and the support for binding allows us to reflect binding in logic directly. We have argued that this system accurately and effectively represents the meta-level of equality, using unknowns and freshness conditions. We studied the proof theory of this logic and we investigated a denotational semantics in nominal sets. In this semantics, object-level variables are first-class entities in the denotation; so an atom a represents an object-level variable symbol in the syntax *and* in the semantics.

From a wider perspective, this paper pursues an overall vision that *names* are worth talking about as mathematical entities, both in logic and in denotation. The use of nominal techniques may be a useful complement to existing methods predicated on function-spaces.

We have seen this in the fact that we have been able to axiomatise *and then study* a representative sample of some of the most influential tools of computer science, including equational logic, first-order logic and substitution. Furthermore, and perhaps most promisingly, we have seen how the flavour of current mathematical systems is maintained in the nominal setting. We are particularly pleased how successfully nominal algebra captures the informal practice of mathematics, in a completely rigorous setting.

8.1 Related Work

8.1.1 Nominal techniques

Nominal algebra has been used to study capture-avoiding substitution for atoms [GM06a, GM07a], and first-order logic [GM06b, GM07c]. Other applications are ongoing.

The derivation rules for freshness and equality (Figures 1 and 2) are inspired by the rules of freshness and α-equivalence from nominal unification [UPG04, Fig. 2 on page 480] and nominal rewriting [FG07, Definition 6 on page 926].

Nominal algebra has similarities to nominal logic [Pit03]. An elementary but important point is that the former is an equational logic whereas the latter is a *first-order* logic with equality. It is not too misleading to imagine that they stand in the same relation as first-order logic and universal algebra — but this intuition is false for an interesting reason; see the discussion of freshness below.

Nominal algebra was used as a basis for nominal equational logic [CP07, page 226], which makes slightly different design decisions. Our rules (**perm**) and (**fr**) correspond to axioms (**F1**) and (**F4**) from [Pit03, page 191], and our rules ($\#\mathbf{X}$) and (**fr**) correspond to the rules ($\not\#$-EQUIVAR) and (ATM-ELIM) from [CP07, Fig. 5 on page 238]. Differences are:

- Nominal equational logic does not use nominal terms, and neither does nominal logic. In nominal equational logic, atoms and abstractions are modelled using families of term-formers, indexed by an infinite collection of atoms. In nominal logic atoms and abstractions are modelled using the sorts of the ambient first-order logic framework.

- The treatment of freshness $a\#t$ is different. Freshness in nominal algebra is designed to model the judgement '$x \notin fv(t)$'. Deciding freshness is syntax directed, always linear in the size of a term, and its validity does not depend on axioms asserting equalities between terms. Therefore $a\#t$ and $t = u$ do not imply that $a\#u$. Consider for example the informal derivation

  $$\lambda x.(((\lambda x.y)x)x) =_\beta \lambda x.(yx) =_\eta y$$

  from Subsection 3.2. Both nominal algebra and nominal equational logic represent this derivation. However, nominal equational logic also represents the following 'incorrect' derivation

  $$\lambda x.(((\lambda x.y)x)x) =_\eta (\lambda x.y)x =_\beta y.$$

  The side-condition $x \notin fv((\lambda x.y)x)$ of the η-equality is not satisfied, but in nominal equational logic freshness is 'semantic' rather than 'syntactic' (as discussed in Subsection 7.3) so we can replace equals with equals; $(\lambda x.y)x$ is equal to y. Derivability of freshness in nominal equational logic is complete with respect to its semantics in nominal sets [CP07, Theorem 10.10], but it may become undecidable if equality of terms is undecidable. Derivability of freshness in nominal algebra is not complete — but nominal algebra expresses semantic freshness in a direct, though unexpected, way using equality (Theorem 7.23), so there is no loss in expressivity. In our opinion the approach to freshness taken by nominal algebra is closer to informal practice.

In summary, nominal logic and nominal equational logic are the first-order and equational logics arising directly from nominal sets (which, viewed as a category, can be presented as the Schanuel Topos [GP02, Mac92]). The treatment of freshness in nominal algebra makes it something a little bit different.

8.1.2 Higher-order techniques

The theory of contexts [Mic01] can be used to axiomatise systems with binding. So, differently, can higher-order algebra [Mei92]. So indeed can simply-typed λ-calculus [Bar00]. These systems are different and intended for different purposes but they share a core which is in essence simply-typed λ-terms up to $\alpha\beta\eta$-equivalence. Just like nominal terms, this richer term-language gives more expressivity which can be used to give stronger axioms. However, meta-variables are represented by *function* variables, and this inherits some distinctive features from their intended functional semantics:

- You have to choose the arity of your unknown in advance. A function variable

  $$F : \overbrace{\mathbb{T} \to \cdots \to \mathbb{T}}^{n} \to \mathbb{T}$$

  can be interpreted as an unknown n-ary predicate — but *which* n? Thus these logics distribute meta-variables across many types.

- Perhaps more importantly, instantiation of these variables avoids capture. This has a side-effect that it is not possible to represent a meta-variable uniformly across contexts. For example to represent a meta-variable ϕ in the context we can intuitively express as $\forall x.\phi$, it suffices to write $\forall\lambda x.F(x)$ where $F : \mathbb{T} \to \mathbb{T}$; but to represent ϕ in the context $\forall x.\forall y.\phi$ we must write $\forall\lambda x.\forall\lambda y.F(x)(y)$ — F must take a higher type. Since for any given F we must choose a type for it, it is not possible to directly represent the context we might write as $\mathsf{Qs}\phi$, where Qs represents an *unknown* context (of quantifiers).

  Instantation of unknowns in nominal algebra does not avoid capture so that $\forall[a]X$ accurately reflects our intention when we write $\forall x.\phi$ where ϕ may be instantiated in a capturing manner. Furthermore X *still* represents ϕ in $\forall[a]\forall[b]X$.

- Moving to higher orders engenders certain computational difficulties. For example unification up to $\alpha\beta$-equivalence is not decidable (although restrictions of it are in [Mil91]), while unification up to CORE is decidable [UPG04].

  Still, quotienting terms by $\alpha\beta\eta$ is convenient, as exploited by theorem-provers such as Isabelle [Pau89]. It remains to be seen whether nominal algebra or something like it can enjoy the success and usefulness enjoyed by Isabelle and similar systems.

In [GJ02, Joj04], Geuvers and Jojgov tackle the first of the above issues by extending higher-order logic with explicit meta-variables. Although the approach of their oHOL language is similar to ours, there are a number of fundamental differences:

- The default notion of instantiation of meta-variables in oHOL is still capture-avoiding; capturing instantiation can be achieved by parameterising the meta-variable, just like in the function variable approach.

- Meta-variables are equipped with pending substitutions of object-variables in oHOL. In nominal algebra meta-variables are equipped with α-renamings of object-variables. Note that we could record these substitutions by using the explicit substitutions from theory SUB.

Besides these fundamental differences, we have not investigated how well nominal algebra can serve as a basis for implementation. From that point of view, Geuvers and Jojgov are far ahead of us.

8.1.3 Functions

Nominal sets provide a model of α-conversion that is pretty close to informal practice. Semantic notions based on functions are models of α-conversion *and* substitution of terms for variables. These are less close to informal practice because they do not have the ability to manipulate names of bound variables *explicitly*. See [GP02, Subsection 1.1] and [Pit03, Section 9] for excellent discussions on this.

However, in addition to α-conversion, we often do need substitution behaviour on nominal sets. We can do this by imposing *axioms* such as theory SUB from Example 2.17. If we can substitute for an atom a then it has the flavour of a variable 'ranging over' denotational elements; so SUB is a theory of 'variables in denotation'. A detailed account of the proof-theory of SUB is in [GM07a].

8.1.4 Binding algebras

Sun's binding algebras [Sun99] are based on a functional semantics for binding, whereas we work according to the relatively newer nominal semantics which is decidedly non-functional; currently the two strands are essentially independent and it remains to see what ideas might flow between them.

Fiore, Plotkin and Turi's binding algebras [FPT99] use categories of presheaves, whereas we use nominal sets. Categories of presheaves do not have a notion of "least supporting set" like nominal sets do [GP02, Related Work]. So in its current form, the freshness judgement of $a\#x$ cannot be expressed in their framework. For this reason, it is not clear how an easy and direct connection can be made between the two frameworks.

8.1.5 Other approaches

A host of 'cylindric' algebraic techniques exist. These embrace meta-variables and reject object-level variables, preferring to encode their expressive power in the term-formers. Examples are lambda-abstraction algebras [Sal00] for the λ-calculus and cylindric algebras [BS81, ANS01] for first-order logic. Combinators [Bar84] reject object-level variables altogether. These systems are effective for their applications, but they cannot *naturally* represent equalities with binding and meta-variables from the simple fact that there are no object-variables.

8.2 Future Work

There is much possible future work.

8.2.1 Axiomatisations of other systems

We have seen examples of three fundamental systems axiomatised in nominal algebra: (capture-avoiding) substitution, the λ-calculus, and first-order logic. In [GM06a, GM07a, GM06b, GM07c] we study the axiomatisations of substitution and first-order logic. We would like to employ the methods of this paper to formalise reasoning on other systems with binding. We speculate on applications to axiomatising substructural logics with quantifiers [Res99]; also to process calculi, some of which feature complex binding side-conditions and for which algebraic reasoning principles are of interest [AG97, Lut02, KD02]; and perhaps even to logics for state such as Bunched Implications and Separation Logic [OP99, Rey02] where atoms (with suitable axioms) could represent locations and freshness might

then nicely express separation. To this end an enriched freshness judgement $t'\#t$ could be useful, generalising $a\#t$ and asserting a separation between the atoms not derivably fresh in t and t'.

Calculi of explicit substitution represent the process of substitution and so represent different aspects of name structure explicitly depending on the particular system [Les94]. A discussion exists of how one might use nominal rewriting to express rewrite systems for λ-calculi with explicit substitutions [FG07, Section 9]; enriching freshness contexts with other judgements would allow reduction strategies to be expressed as part of the rewrite rules. It would also be interesting to attempt nominal algebra or nominal rewriting axiomatisations of λ-calculi with non-standard treatments of binding, such as Adbmal [HvO03].

8.2.2 Semantics

Although we have specified a sound and complete semantics for nominal algebra theories in nominal sets, we have not yet proved the HSP theorem (also known as Birkhoff's theorem) [BS81]. Freshness contexts introduce a flavour of implication into nominal algebra axioms, so it is not a priori obvious that the HSP theorem survives this extension though the authors believe that it does. The HSP theorem is one of the basic results of universal algebra. It gives a valuable insight into the structure of models of algebraic theories, so this is important future work.

8.2.3 Hierarchies of variables

We are also interested in developing logics with *hierarchies* of 'increasingly meta-'variables. Since nominal algebra offers two levels of variable, why not extend this to allow an infinite hierarchy of variables, by analogy with type hierarchies in the λ-calculus [Bar84]? Work already started in this direction by extending nominal terms with a hierarchy of variables [Gab05, Gab07b, GL07]. We would like to extend these hierarchical nominal terms to a logic framework of equations.

8.2.4 Implementation

We interested in exploring how well nominal algebra could serve as the basis for an implementation of an interactive proof assistent. This may be useful because, as discussed in the Introduction, nominal algebra permits a particularly direct translation between informal mathematical practice and formal syntax; also, systems based on nominal techniques, such as nominal unification and rewriting [UPG04, FG07], may have more favourable algorithmic properties than the current state of the art based on higher-order techniques (syntax up to $\alpha\beta$-equivalence).

References

[AG97] Martín Abadi and Andrew D. Gordon. A calculus for cryptographic protocols: the spi calculus. In *CCS '97: Proc. of the 4th ACM conf. on Computer and Communications Security*, pages 36–47. ACM Press, 1997.

[ANS01] H. Andréka, I. Németi, and I. Sain. Algebraic logic. In D.M. Gabbay and F. Guenthner, editors, *Handbook of Philosophical Logic, 2nd Edition*, volume 2, pages 133–249. Kluwer, 2001.

[Bar75] Jon Barwise. *Admissible Sets and Structures: an approach to definability theory*. Perspectives in mathematical logic. Springer, 1975.

[Bar84] H.P. Barendregt. *The Lambda Calculus: its Syntax and Semantics (revised ed.)*. North-Holland, 1984.

[Bar00] H.P. Barendregt. Lambda calculi with types. In S. Abramsky, D. M. Gabbay, and T. S. E. Maibaum, editors, *Handbook of Logic in Computer Science, Volume 2*, pages 117–309. OUP, 2000.

[Bru96] N. Brunner. 75 years of independence proofs by Fraenkel-Mostowski permutation models. *Mathematica Japonica*, 43(1):177–199, 1996.

[BS81] S. Burris and H. Sankappanavar. *A Course in Universal Algebra*. Springer, 1981.

[CF58] Haskell B. Curry and Robert Feys. *Combinatory Logic*, volume 1. North Holland, 1958.

[CP07] Ranald A. Clouston and Andrew M. Pitts. Nominal equational logic. *ENTCS*, 172:223–257, 2007.

[dB91] N.G. de Bruijn. Checking mathematics with computer assistance. *Notices of the American Mathematical Society (AMS)*, 38(1):8–15, 1991.

[FG07] Maribel Fernández and Murdoch J. Gabbay. Nominal rewriting. *Information and Computation*, 205:917–965, 2007.

[FPT99] Marcelo Fiore, Gordon Plotkin, and Daniele Turi. Abstract syntax and variable binding. In *14th Annual Symposium on Logic in Computer Science*, pages 193–202. IEEE Computer Society Press, 1999.

[Gab05] Murdoch J. Gabbay. A new calculus of contexts. In *PPDP '05: Proc. of the 7th ACM SIGPLAN international conference on Principles and Practice of Declarative Programming*, pages 94–105. ACM Press, 2005.

[Gab06] Murdoch J. Gabbay. Fresh logic. *Journal of Logic and Computation*, 2006. In press.

[Gab07a] Murdoch J. Gabbay. A general mathematics of names. *Information and Computation*, 205(7):982–1011, 2007.

[Gab07b] Murdoch J. Gabbay. Hierarchical nominal terms and their theory of rewriting. *ENTCS*, 174(5):37–52, 2007.

[GJ02] Herman Geuvers and Gueorgui I. Jojgov. Open proofs and open terms: A basis for interactive logic. In *Computer Science Logic: 16th International Workshop*, pages 537–552, 2002.

[GL07] Murdoch J. Gabbay and Stéphane Lengrand. The lambda-context calculus. In *LFMTP'07: International Workshop on Logical Frameworks and Meta-Languages*, to appear in ENTCS, 2007.

[GM06a] Murdoch J. Gabbay and Aad Mathijssen. Capture-avoiding substitution as a nominal algebra. In *ICTAC'2006: 3rd International Colloquium on Theoretical Aspects of Computing*, volume 4281 of *LNCS*, pages 198–212, 2006.

[GM06b] Murdoch J. Gabbay and Aad Mathijssen. One-and-a-halfth-order logic. In *PPDP '06: Proc. of the 8th ACM SIGPLAN symposium on Principles and Practice of Declarative Programming*, pages 189–200. ACM Press, 2006.

[GM07a] Murdoch J. Gabbay and Aad Mathijssen. Capture-avoiding substitution as a nominal algebra. *Formal Aspects of Computing*, 2007. To appear.

[GM07b] Murdoch J. Gabbay and Aad Mathijssen. A formal calculus for informal equality with binding. In *WoLLIC'07: 14th Workshop on Logic, Language, Information and Computation*, volume 4576 of *LNCS*, pages 162–176, 2007.

[GM07c] Murdoch J. Gabbay and Aad Mathijssen. One-and-a-halfth-order logic. *Journal of Logic and Computation*, 2007. To appear.

[GMR+07] Jan Friso Groote, Aad Mathijssen, Michel Reniers, Yaroslav Usenko, and Muck van Weerdenburg. The formal specification language mCRL2. In *Methods for Modelling Software Systems (MMOSS)*, number 06351 in Dagstuhl Seminar Proceedings. Internationales Begegnungs- und Forschungszentrum fuer Informatik (IBFI), Schloss Dagstuhl, Germany, 2007.

[GP02] Murdoch J. Gabbay and Andrew M. Pitts. A new approach to abstract syntax with variable binding. *Formal Aspects of Computing*, 13(3–5):341–363, 2002.

[Gro97] Jan Friso Groote. The syntax and semantics of timed μCRL. Technical Report SEN-R9709, CWI, Amsterdam, 1997.

[HMT85] L. Henkin, J.D. Monk, and A. Tarski. *Cylindric Algebras*. North Holland, 1971 and 1985. Parts I and II.

[HvO03] Dimitri Hendriks and Vincent van Oostrom. Adbmal. In *CADE*, pages 136–150, 2003.

[Joh87] P.T. Johnstone. *Notes on logic and set theory*. Cambridge University Press, 1987.

[Joj04] Gueorgui I. Jojgov. *Incomplete Proofs and Terms and Their Use in Interactive Theorem Proving*. PhD thesis, Technische Universiteit Eindhoven, 2004.

[KD02] Joost-Pieter Katoen and Pedro R. D'Argenio. General distributions in process algebra. In *Lectures on formal methods and performance analysis: first EEF/Euro summer school on trends in computer science*, pages 375–429. Springer, 2002.

[KKSdV97] J.R. Kennaway, J.W. Klop, M.R. Sleep, and F.J. de Vries. Infinitary lambda calculus. *Theoretical Computer Science*, 175:93–125, 1997.

[Les94] Pierre Lescanne. From lambda-sigma to lambda-upsilon a journey through calculi of explicit substitutions. In *POPL '94: Proc. 21st ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages*, pages 60–69. ACM Press, 1994.

[LS04] Stefania Lusin and Antonino Salibra. The lattice of lambda theories. *Journal of Logic and Computation*, 14(3):373–394, 2004.

[Lut02] Bas Luttik. *Choice Quantification in Process Algebra*. PhD thesis, University of Amsterdam, 2002.

[Mac92] Mac Lane and Moerdijk. *Sheaves in Geometry and Logic: A First Introduction to Topos Theory*. Universitext. Springer, 1992.

[Mei92] Karl Meinke. Universal algebra in higher types. *Theoretical Computer Science*, 100(2):385–417, 1992.

[Mic01] Marino Miculan. Developing (meta)theory of lambda-calculus in the theory of contexts. *ENTCS*, 58(1), 2001.

[Mil91] Dale Miller. A logic programming language with lambda-abstraction, function variables, and simple unification. *Extensions of Logic Programming*, 475:253–281, 1991.

[OP99] Peter W. O'Hearn and David J. Pym. The logic of bunched implications. *Bulletin of Symbolic Logic*, 5(2):215–244, 1999.

[Par01] Joachim Parrow. An introduction to the pi-calculus. In Jan Bergstra, Alban Ponse, and Scott Smolka, editors, *Handbook of Process Algebra*, pages 479–543. Elsevier Science, 2001.

[Pau89] Lawrence C. Paulson. The foundation of a generic theorem prover. *Journal of Automated Reasoning*, 5(3):363–397, 1989.

[Pit03] Andrew M. Pitts. Nominal logic, a first order theory of names and binding. *Information and Computation*, 186(2):165–193, 2003.

[Res99] Greg Restall. *An Introduction to Substructural Logics*. Routledge, 1999.

[Rey02] John C. Reynolds. Separation logic: A logic for shared mutable data structures. In *LICS*, pages 55–74. IEEE Computer Society, 2002.

[Sal00] Antonino Salibra. On the algebraic models of lambda calculus. *Theoretical Computer Science*, 249(1):197–240, 2000.

[Sun99] Yong Sun. An algebraic generalization of Frege structures - binding algebras. *Theoretical Computer Science*, 211:189–232, 1999.

[UPG04] Christian Urban, Andrew M. Pitts, and Murdoch J. Gabbay. Nominal unification. *Theoretical Computer Science*, 323(1–3):473–497, 2004.

[vB01] Johan van Benthem. Higher-order logic. In D.M. Gabbay and F. Guenthner, editors, *Handbook of Philosophical Logic, 2nd Edition*, volume 1, pages 189–244. Kluwer, 2001.

A ZFA Equivariance

We use atoms in this paper — we introduce them when we write 'Fix a countably infinite collection of **atoms** $a, b, c, \ldots$' in Definition 2.1.

Atoms have a very special property: We can tell them apart — $a \neq b$ is true and $a = b$ is false — but we cannot compare them in any other way — for example there is no order $a < b$. This distinguishes atoms from, say, the natural numbers, set ordinals, and so on.

If we bear this property of atoms in mind we obtain equivariance. This is a meta-mathematical property; it refers to the assertions written *in English in this paper* about nominal algebra, equational logic and so on.

Definition A.1. The language of ZFA set theory is first-order logic with equality with in addition:

(**Sets**)	$\forall x.((\exists y.y \in x) \supset x \notin \mathbb{A})$
(**Extensionality**)	$\forall x.(x \notin \mathbb{A} \supset x = \{z \mid z \in x\})$
(**Comprehension**)	$\forall x.\exists y.(y \notin \mathbb{A} \wedge y = \{z \in x \mid \phi\})$ (y not free in ϕ)
($\in$-**Induction**)	$\forall x.(\forall y.(y \in x \supset \phi[y/x]) \supset \phi) \supset \forall x.\phi$
(**Replacement**)	$\forall x.\exists z.(z \notin \mathbb{A} \wedge z = \{F(y) \mid y \in x\})$
(**Pairset**)	$\forall x.\forall y.\exists z.(z = \{x, y\})$
(**Union**)	$\forall x.\exists z.(z \notin \mathbb{A} \wedge z = \{y \mid \exists y'.(y \in y' \wedge y' \in x)\})$
(**Powerset**)	$\forall x.\exists z.(z = \{y \mid y \subseteq x\})$
(**Infinity**)	$\exists x.(\emptyset \in x \wedge \forall y.(y \in x \supset y \cup \{y\} \in x))$

Figure 4: Axioms of ZFA set theory

- A binary predicate symbol $\in$ called *set membership*.
- A constant term-former $\mathbb{A}$ called *the set of atoms*.

We use standard sugar of classical logic.

Definition A.2. ZFA set theory has the axioms in Figure 4.

In Figure 4, ϕ ranges over all predicates, $\phi[y/x]$ denotes the predicate obtained by capture-avoiding substitution of x by y, and $F(y)$ represents any function which can be expressed in the language of ZFA sets. We also use the following sugar:

$x = \{z \mid z \in x\}$	is sugar for	$\forall y.(\forall z.(z \in x \Leftrightarrow z \in y) \supset x = y)$
$y = \{z \in x \mid \phi\}$	is sugar for	$\forall z.(z \in y \Leftrightarrow (z \in x \wedge \phi))$
$z = \{F(y) \mid y \in x\}$	is sugar for	$\forall u.(u \in z \Leftrightarrow \exists y.(F(y) = u \wedge y \in x))$
$z = \{x, y\}$	is sugar for	$\forall u.(u \in z \Leftrightarrow (u = x \vee u = y))$
$z = \{y \mid \exists y'.(y \in y' \wedge y' \in x)\}$	is sugar for	$\forall y.(y \in z \Leftrightarrow \exists y'.(y \in y' \wedge y' \in x))$
$z = \{y \mid y \subseteq x\}$	is sugar for	$\forall y.(y \in z \Leftrightarrow \forall y'.(y' \in y \supset y' \in x))$
$\emptyset \in x$	is sugar for	$\exists z.(z \in x \wedge \forall z'.z' \notin z)$
$y \cup \{z\} \in x$	is sugar for	$\exists u.(u \in x \wedge \forall u'.(u' \in u \Leftrightarrow u \in y \vee u = z))$

The syntactic sugar used in set theory is very rich; further details can be found elsewhere [Joh87]. The only property we care about is that set theory is a foundational theory and is rich enough to express, in principle at least, all the mathematics in the rest of this paper.

Definition A.3. We define a **permutation action** on ZFA sets by:

$$\pi \cdot a = \pi(a) \qquad \pi \cdot X = \{\pi \cdot x \mid x \in X\} \quad (X \notin \mathbb{A})$$

This definition is by ϵ-induction, a standard method in set theory [Joh87] which relies on a well-foundedness property implied by ($\in$-**Induction**).

Recall that ϕ ranges over predicates of ZFA. Write $\phi(x_1, \ldots, x_n)$ to range over predicates which mention at most $x_1, \ldots, x_n$ as free variable symbols.

Theorem A.4 (ZFA equivariance)**.** *If $\phi(x_1, \ldots, x_n)$ is a predicate of ZFA set theory then*

$$\phi(x_1, \ldots, x_n) \Leftrightarrow \phi(\pi \cdot x_1, \ldots, \pi \cdot x_n)$$

is always provable.

As a corollary, $\phi(x_1, \ldots, x_n)$ and $\phi(\pi \cdot x_1, \ldots, \pi \cdot x_n)$ are interchangeable in proof and in validity on models.

Proof. We work by induction on the syntax of ϕ.

- By definition, $x \in y$ implies $\pi \cdot x \in \pi \cdot y$ follows directly from the fact that $\pi \cdot y = \{\pi \cdot y' \mid y' \in y\}$. The reverse implication is easy using π^{-1}.
- Similarly, $x = y$ if and only if $\pi \cdot x = \pi \cdot y$.
- The case of $\bot$ is trivial, and the cases of $\phi_1 \supset \phi_2$ and $\forall z.\phi'$ follow using the inductive hypothesis.
- $\pi \cdot \mathbb{A} = \mathbb{A}$ is provable, so $x \in \mathbb{A}$ if and only if $\pi \cdot x \in \mathbb{A}$, and $\mathbb{A} \in y$ if and only if $\mathbb{A} \in \pi \cdot y$, and similarly $x = \mathbb{A}$ if and only if $\pi \cdot x = \mathbb{A}$ and $\mathbb{A} = y$ if and only if $\mathbb{A} = \pi \cdot y$.

The result follows. □

Equivariance was first observed by Fraenkel and Mostowski and used to prove the independence of the axiom of choice from the other axioms of set theory [Bru96]. Atoms are atomic objects with no internal structure, so it is natural to use these to model variable symbols. This idea appears already in [Bar75]. To our knowledge the first author's PhD thesis [GP02] observed the application of equivariance to practical reasoning on variables in computer science as we use it in this paper.

We have used equivariance freely in structural inductive proofs to freely rename atoms in inductive hypotheses where convenient, while remaining fully formal; this technique was first used as part of a proof of cut-elimination for a sequent system for nominal logic [Gab06]. If we wish to be fully formal but ignore equivariance then we must work by induction on measures such as term length or derivation depth. These proofs tend to be longer, harder to read — and they are rarely given in full detail outside of a theorem-prover.